

ISCB GMDS 2026

BIG DATA
small data
Your Data

Transitions in the age of biomedical AI

FREIBURG | GERMANY

27.09. - 01.10.26

Pre-conference course
**Clinical Text-Mining under
Real-World Constraints**

27.09.2026 | Vittorio Torri, Francesca Ieva

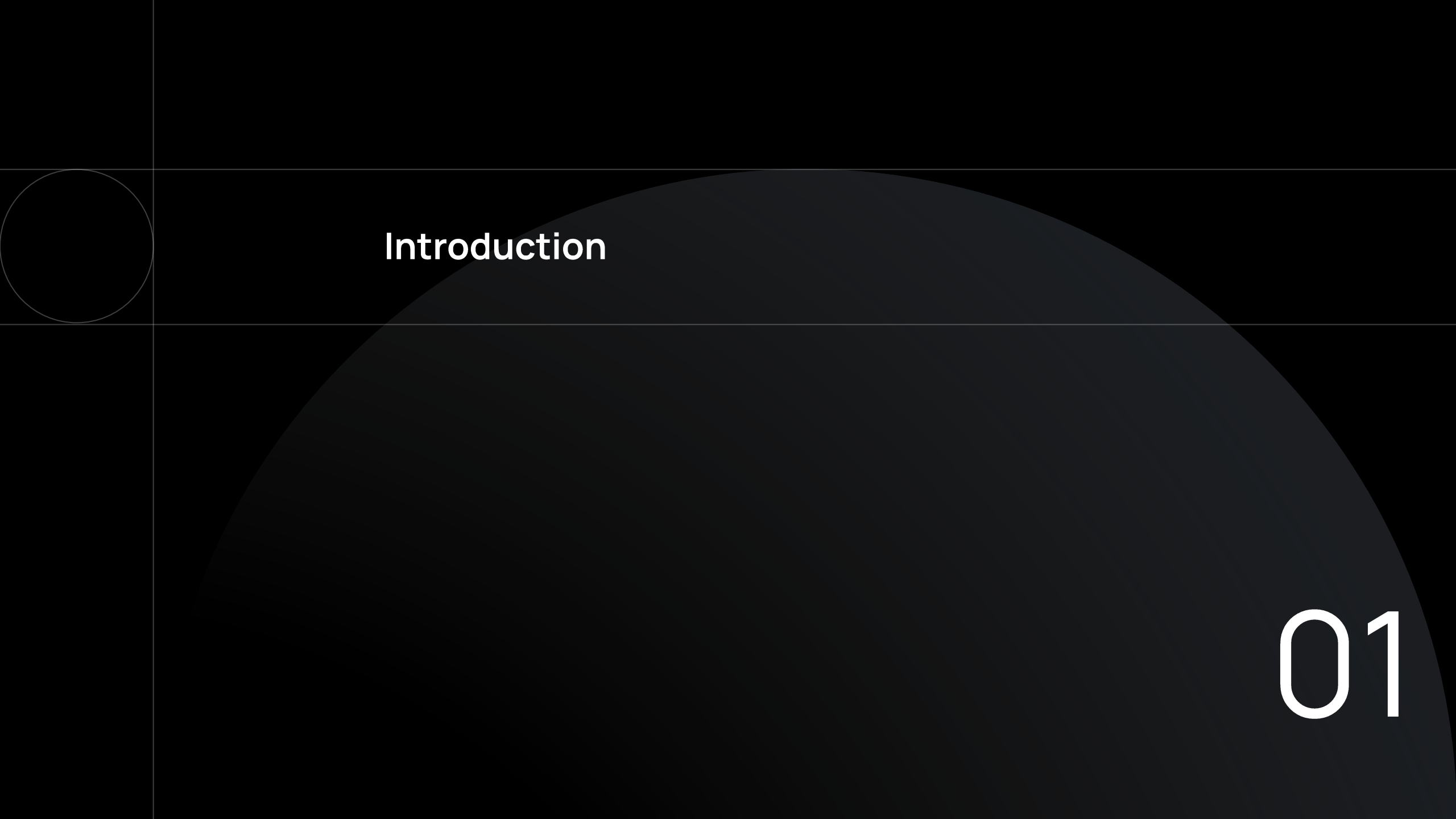


POLITECNICO
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DIPARTIMENTO DI MATEMATICA
DEPARTMENT OF EXCELLENCE 2023-2027

Agenda

1. Introduction
2. Natural Language Processing 101
3. Clinical NLP: Challenges and Opportunities
4. Clustering short clinical texts
5. Weakly-supervised classification of clinical documents
6. Interpretability methods for document classification
7. LLM-prompting for information extraction



Introduction

01

Who we are



VITTORIO TORRI

Postdoctoral Researcher

Health Analytics group @ MOX Lab, Department of Mathematics, Politecnico di Milano

His research focuses on NLP for clinical documents, focusing on methods to counter label and resource scarcity and to improve interpretability of black-box models. He has worked on multiple NLP-related projects in collaboration with Italian and Dutch hospitals and public health institutions.



FRANCESCA IEVA

Associate Professor of Statistics

Health Analytics group @ MOX Lab, Department of Mathematics, Politecnico di Milano

Associate Head

Health Data Science Centre, Human Technopole

Her research spans statistical learning for biomedical data, from hierarchical, multi-state models and functional data analysis to more recent AI methods for genomics, medical images, and clinical documents.

What we aim to do today



Provide a minimal **theoretical background** on the analysis of textual data

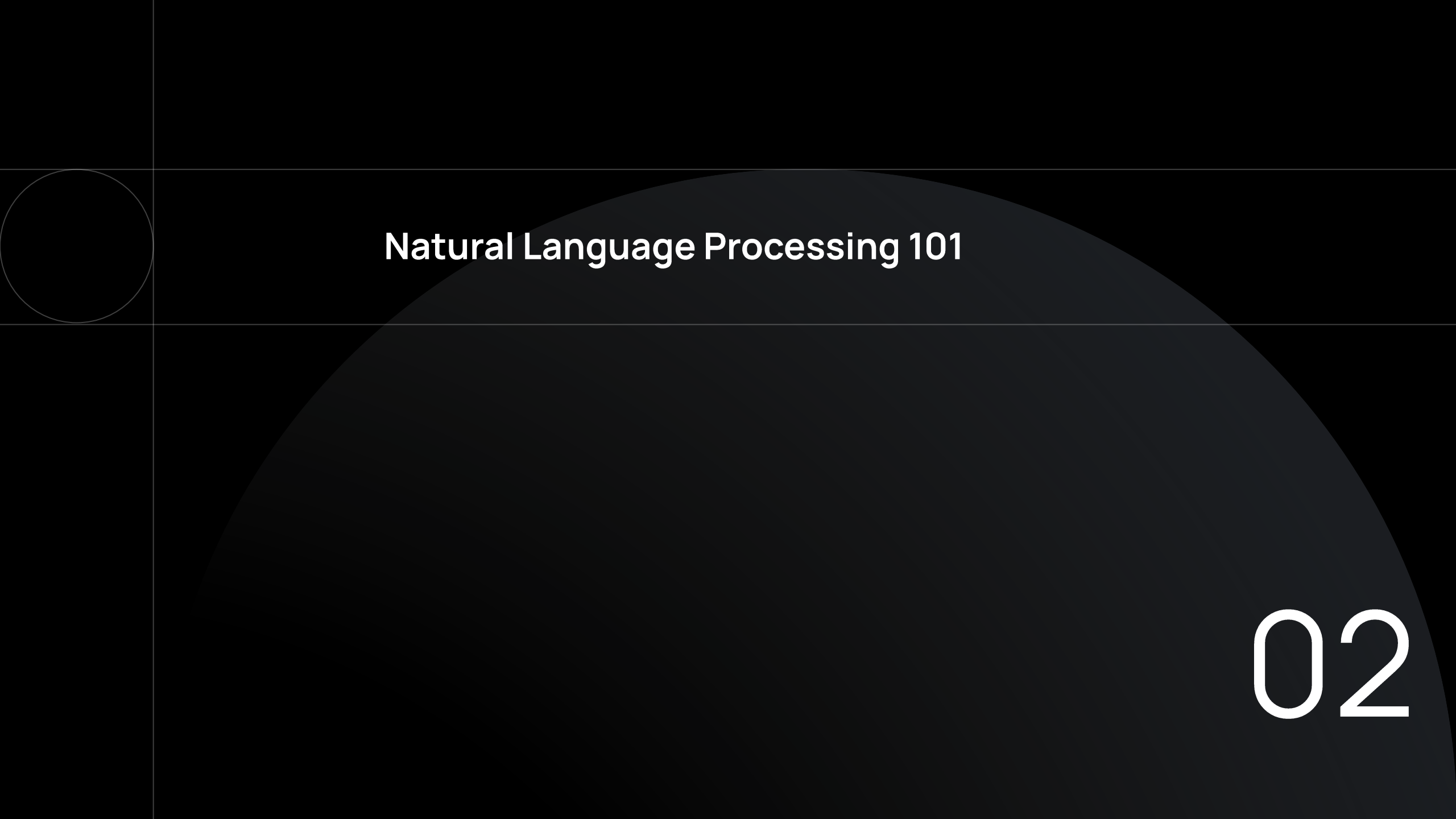


Discuss **opportunities** and **challenge** of **clinical** textual data



Hands-on, interactive examples, derived from real-world case studies, of methods and techniques for extracting knowledge from medical documents

Who are you?



Natural Language Processing 101

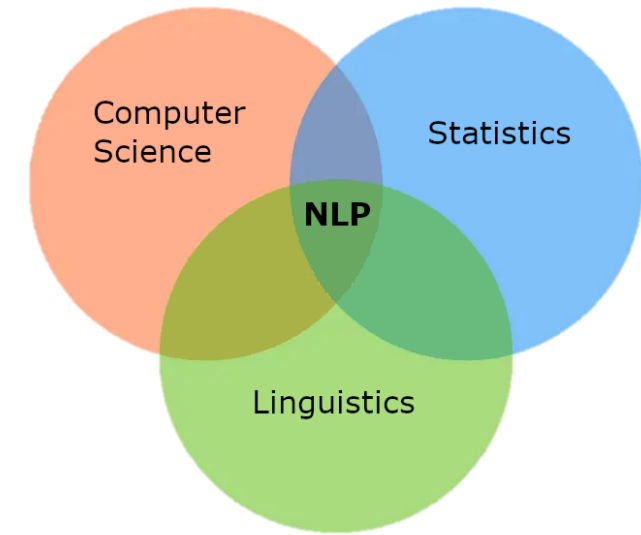
02

NLP 101 – What is Natural Language Processing?

Language: A system of communication consisting of sounds, words, and grammar

Natural Language: A language that has evolved naturally as a means of communication among people, in contrast with artificial/formal languages such as programming languages

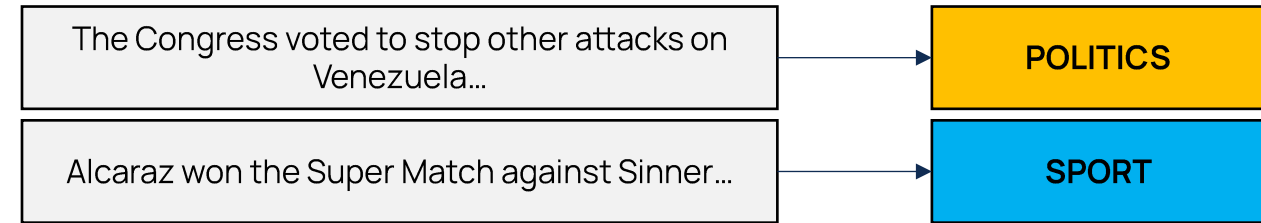
Natural Language Processing (NLP): An area of research at the intersection of computer science, statistics and linguistics that involves giving computers and digital devices the ability to recognize, understand and generate language



NLP 101 – Tasks (I)

Classification

Recognize topics of news, sentiment from customer reviews, diagnoses from health documents



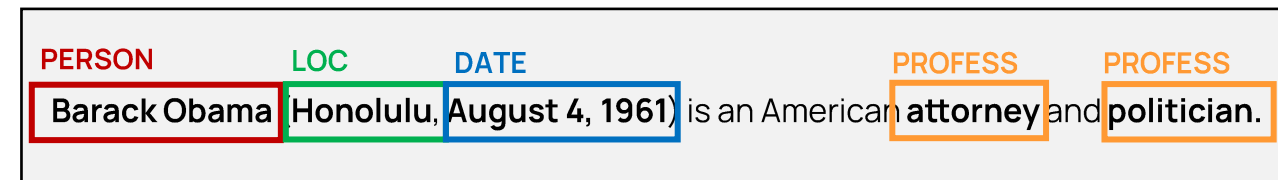
Regression

Sentiment score from customer reviews, target price from house description, risk score from patients' documents



Named Entity Recognition

Medication and procedures from health documents, people and assets in legal documents



Entity Linking and/or Normalization

Link entities to known ontologies (ex: UMLS) or transform into standard formats (ex: dates, addresses)

Barack Obama → dbr:Barack-Obama
August 4, 1961 → Day: 4 Month: 8 Year: 1961
USA → dbr:Honolulu

NLP 101 – Tasks (II)

- **Clustering (Topic Modeling)**

Unsupervised diagnoses identification from corpora of medical documents, analysis of social media posts

Japanese company is introducing cats in its offices..

Wolves are coming back on the Italian Alps...

The Federal Reserve is keeping interest rates

NASDAQ today fell due to worries about supplies...

- **Text generation**

Question answering, summarization, generation of synthetic documents

Q. When was Franklin D. Roosevelt born?

A. President Franklin D. Roosevelt was born in January 1882

NLP 101 – The fundamental problem of NLP

Mathematical models work with **numbers**

Structured data are numbers or can be easily converted into numbers

Images can be (relatively easily) represented as numbers

But it is not immediate to convert text into (meaningful) numbers!

time	bmi	anaemia_1	diabetes_1	high_blood_pressure_1	sex_Male
4	30.82	0.00	0.00	1.00	1.00
6	25.40	0.00	0.00	0.00	1.00
7	29.45	0.00	0.00	0.00	1.00
7	36.06	1.00	0.00	0.00	1.00
8	33.69	1.00	1.00	0.00	0.00

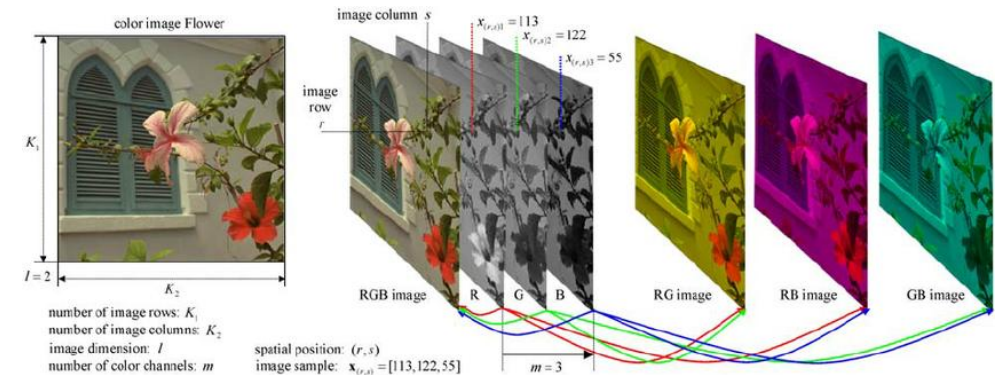


Figure from doi:10.1201/9781420009781.ch16

NLP 101 – Tokenization (I)

A text is a sequence of **words**.

Every type of language modelling requires to split somehow the text into smaller elements: **tokens**

Many questions arise with tokenization and many solutions have been proposed

In **New York**, the engineering team at **Hewlett-Packard** spent days debating the **model's** ability to interpret complex texts with **remarkable** clarity.

- «New York» or «New» + «York» ?
- «Hewlett-Packard» or «Hewlett» + «Packard» ?
- «model», «models» or «model» + «s» ?
- «remarkable» or «remark» or «remark» + «able» ?

NLP 101 – Tokenization (II)

Word-level

- Interpretable
- Simple
- High risk of Out Of Vocabulary (OOV) words
- Loses meaningful relationships among different word forms

[In] [New] [York] [,] [the]
[engineering] [team] [at] [Hewlett-
Packard] ...

Character-level

- No OOV issues
- Robust to casing and misspelling errors
- Too slow
- Loses the meaning

[l] [n] [N] [e] [w] [Y] [o] [r] [k] [,] [t]
[h] [e] [e] [n] [g] [i] [n] [e] [e] [r] [i]
[n] [g] ...

Sub-word level

- Trade-off between word and characters
- Can exploit knowledge about prefixes and suffixes
- Limited OOV risk

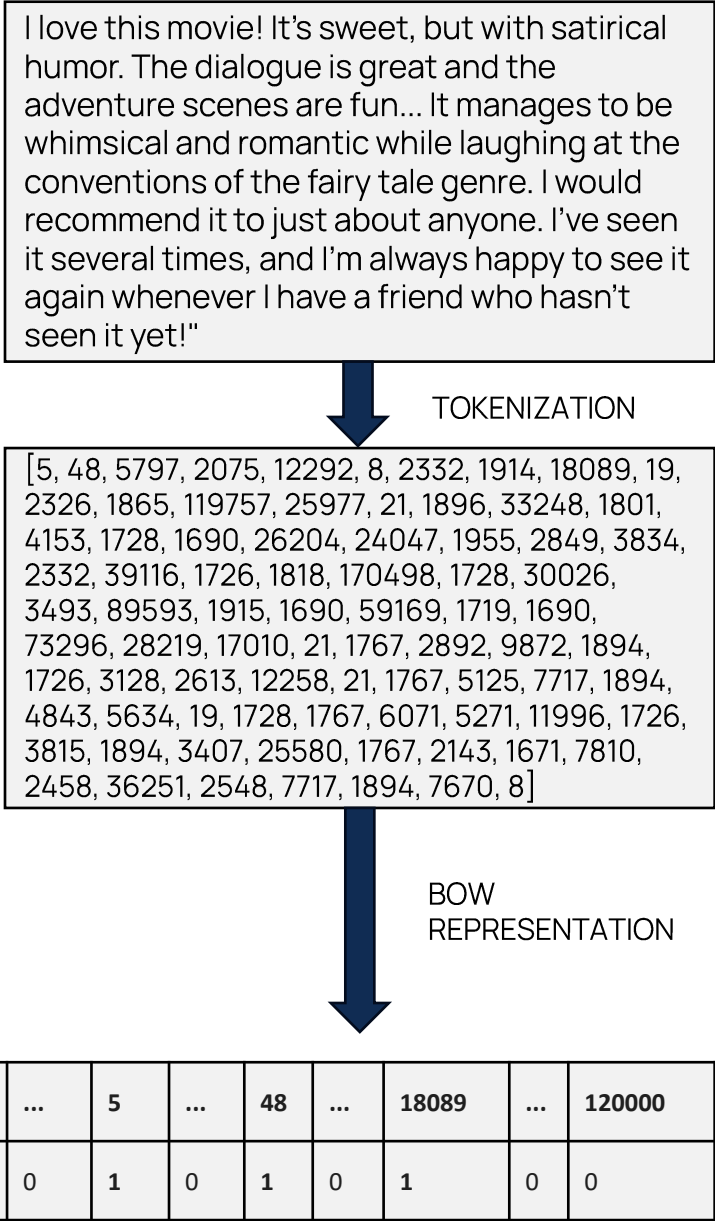
[In] [ĠNew] [ĠYork] [,] [Ġthe]
[Ġengine] [ering] [Ġteam] [Ġat]
[ĠHewl] [ett] [-] [Pack] [ard] ...

NLP 101 – Bag of words representation

After tokenization, how do we feed the tokenized text as input for a model?

The simplest option, named **Bag of Words**, act similarly to One Hot Encoding:

- Each document is converted into a **vector**
- Size of vector = **# words in the vocabulary**
- Position **#i** corresponds to words **#i** in the vocabulary and its value can be:
 - **0** if the i-th word is **not present** in the doc or **1** if the i-th word **is present** in the text
 - **0** if the i-th word is **not present** in the doc or **n** if the i-th word has **n occurrences** in the text



NLP 101 – TF-IDF representation

Variations of Bag of Words use weighted values

The most common is **TF-IDF**:

$TF(t, d) = \# \text{ occurrences of token } t \text{ in document } d$

$df(t) = |d \in D : t \in d|$

$IDF(t) = \log \frac{N}{1 + df(t)}$

$TFIDF(t, d) = TF(t, d) \cdot IDF(t)$

TOKEN_ID	TERM	TF	df	IDF	TF-IDF
1688	it	6	82	0,08	0,49
48	I	5	51	0,28	1,42
3798	the	4	89	0,05	0,18
1726	to	3	61	0,21	0,62
1728	and	2	76	0,11	0,23
7717	seen	1	22	0,64	0,64
7670	yet	1	10	0,96	0,96
82015	would	1	12	0,89	0,89
131587	whimsical	1	1	1,70	1,70
31232	times	1	15	0,80	0,80
10304	weet	1	6	1,15	1,15
45949	satirical	1	20	0,68	0,68
128843	adventure	1	25	0,59	0,59
37563	genre	1	30	0,51	0,51
38081	fairy	1	14	0,82	0,82
20716	humor	1	35	0,44	0,44
33258	have	1	62	0,20	0,20
60668	great	1	41	0,38	0,38



Token ID	0	...	48	...	1726	...	1728	...	...
"I love..."	0	0	1.2	0	0.62	0	0.23	0	...

NLP 101 – Vocabulary-based representation issues (I)

Size of vocabulary can be huge: **Heap's law:** $V(n) \propto n^\beta$ $\beta \in (0,4; 0,6)$ with n documents

→ need to apply a **strong feature selection**:

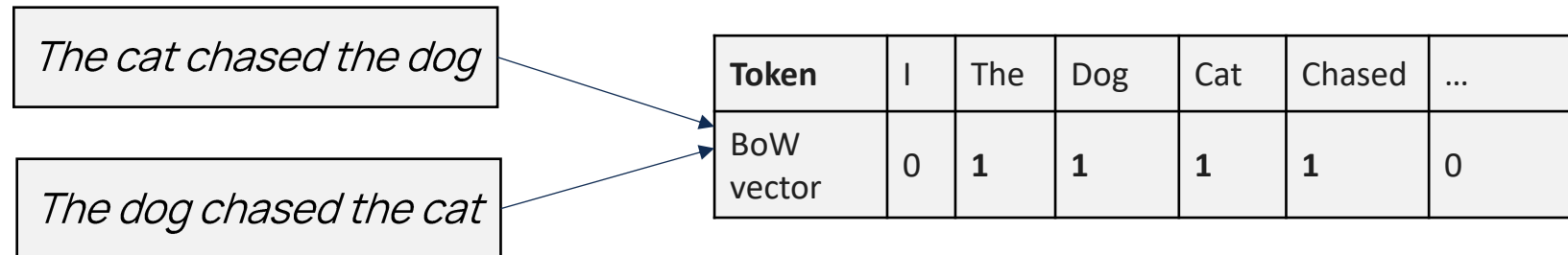
- Delete the **stopwords**: *propositions, articles, and conjunctions* that do not convey much information but sometimes these words are useful and/or necessary!
- Delete words with very low or very high frequency
- **Stemmatization**: remove standard suffixes from words
- **Lemmatization**: replace each word with its root form (language-specific morphological parsers)

*he **is** reading detective stories → he **be** read detective story*

NLP 101 – Vocabulary-based representation issues (II)

STRUCTURAL BLINDNESS

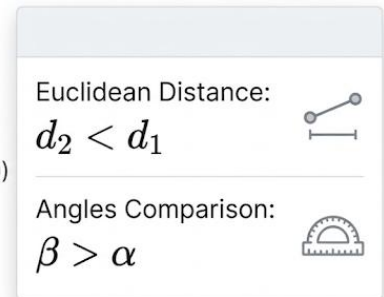
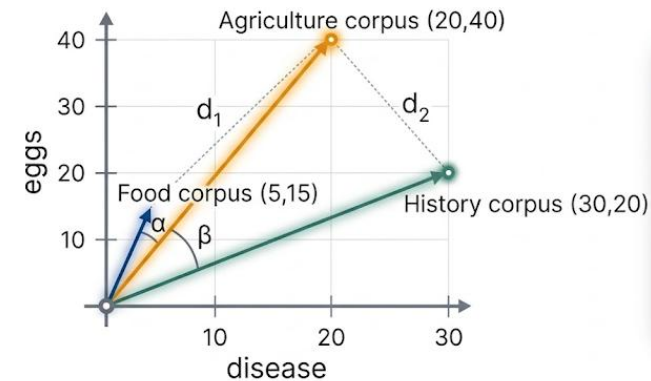
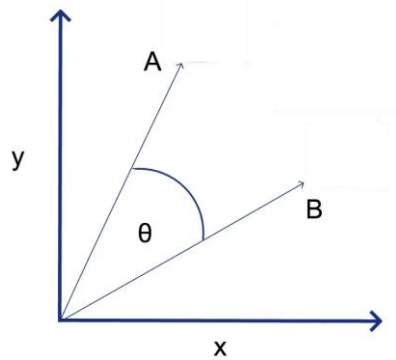
Vocabulary-based representations treat text as a set, not as a sequence



NLP 101 – Vocabulary-based representation issues (III)

SEMANTIC GAP

- This representation does not account for similarity between words (ex: *synonyms*)
- Length of documents can alter similarity → use **cosine similarity** to avoid this



$$\text{cosine_similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2 \sum_{i=1}^n B_i^2}}$$

NLP 101 – Static Word Embeddings

“The meaning of a word is its use in the language” (L. Wittgenstein)

A completely different type of text representation are **embeddings** produced by **neural networks**

Word2Vec (Mikolov, 2013) was the first notable example, followed by Glove (Pennington, 2014)

KEY IDEA

Learn **dense, fixed size, continuous** word vectors exploiting a proxy task that allows to capture the **semantic relationships** existing among words



Mikolov, Tomas et al. “Efficient Estimation of Word Representations in Vector Space.” *International Conference on Learning Representations (2013)*.

Bengio, Yoshua, et al. “A neural probabilistic language model.” *Journal of Machine Learning Research* 3.Feb (2003): 1137-1155.

Pennington, Jeffrey, Richard Socher, and Christopher D. Manning. “Glove: Global vectors for word representation.” *Proceedings of the 2014 EMNLP Conference*. 2014.

NLP 101 – Word2Vec

Word2Vec can be implemented with different architectures.

The simplest is the **Continuous Bag-of-Word (CBOW)**:

A one-hidden-layer feed-forward neural network

Input: n words (one-hot encoded) of a sentence in which one word in the middle has been removed

Output: the missing word (softmax over vocabulary)

The matrix W which captures the embeddings is shared among the input words

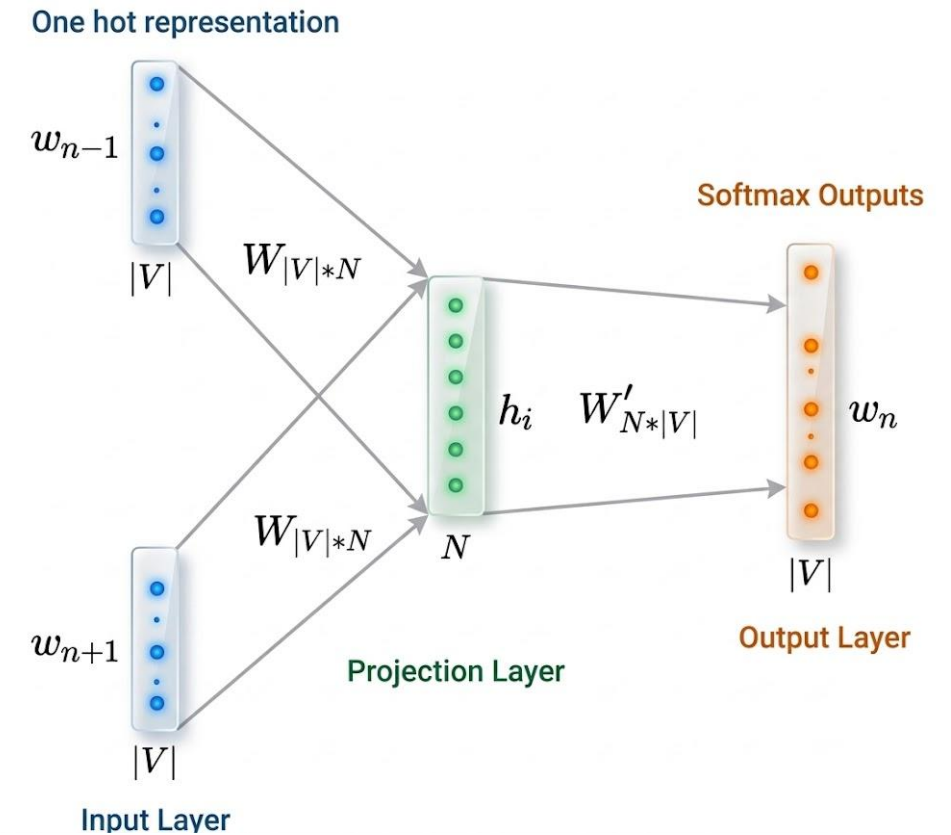
$$W \in \mathcal{R}^{|V| \times N}$$

$|V|$ = size of the vocabulary

N = size of the embeddings (hidden layer size)

Unsupervised training on a (large) corpus

After training, W is an embedding lookup table



NLP 101 – Semantic relationships in static word embeddings

Static word embeddings are able to learn relevant semantic and syntactic relationships among words

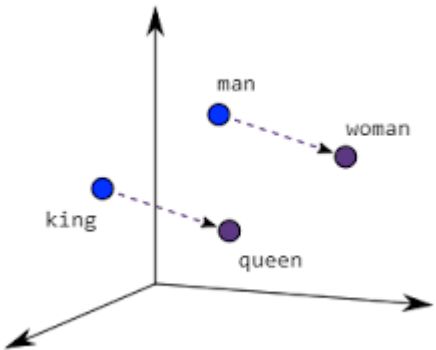
Examples of interesting relationships learnt by Word2Vec:

Relationship	Example 1	Example 2	Example 3
France - Paris	Italy: Rome	Japan: Tokyo	Florida: Tallahassee
big - bigger	small: larger	cold: colder	quick: quicker
Miami - Florida	Baltimore: Maryland	Dallas: Texas	Kona: Hawaii
Einstein - scientist	Messi: midfielder	Mozart: violinist	Picasso: painter
Sarkozy - France	Berlusconi: Italy	Merkel: Germany	Koizumi: Japan
copper - Cu	zinc: Zn	gold: Au	uranium: plutonium
Berlusconi - Silvio	Sarkozy: Nicolas	Putin: Medvedev	Obama: Barack
Microsoft - Windows	Google: Android	IBM: Linux	Apple: iPhone
Microsoft - Ballmer	Google: Yahoo	IBM: McNealy	Apple: Jobs
Japan - sushi	Germany: bratwurst	France: tapas	USA: pizza

$A:B s.t. B-A=R$

$C:D s.t. C+R=D$

Relationship vector R between two words (A and B) is obtained by subtracting word vector B from word vector A. For each example, the first word vector C is summed with R, and the word vector D which is the nearest to C+R holds a similar relationship (C,D) as the original (A,B)



NLP 101 – Static embeddings issues

POLYSEMY

I deposited money at the *bank*

VS

The boat is stuck on the river *bank*

bank has always the same embedding

STRUCTURAL BLINDNESS

The dog chase the cat

VS

The cat chase the god

The sentence embedding is the same

NLP 101 – Transformers (I)

*I arrived at the **bank** after crossing the ... street? ... river?*

To capture the **contextual meaning** of a word we need to look at all surrounding words

but the meaning of a word (ex: **bank**) is often unclear until the end of the sentence, requiring a slow sequential processing (RNN architectures)

Transformer is an architecture that avoid this sequential processing using a mechanism called **self-attention** (+ **positional encoding**), which requires a constant number of steps to process any sentence, independently from its length.

Animation from <https://research.google/blog/transformer-a-novel-neural-network-architecture-for-language-understanding/>

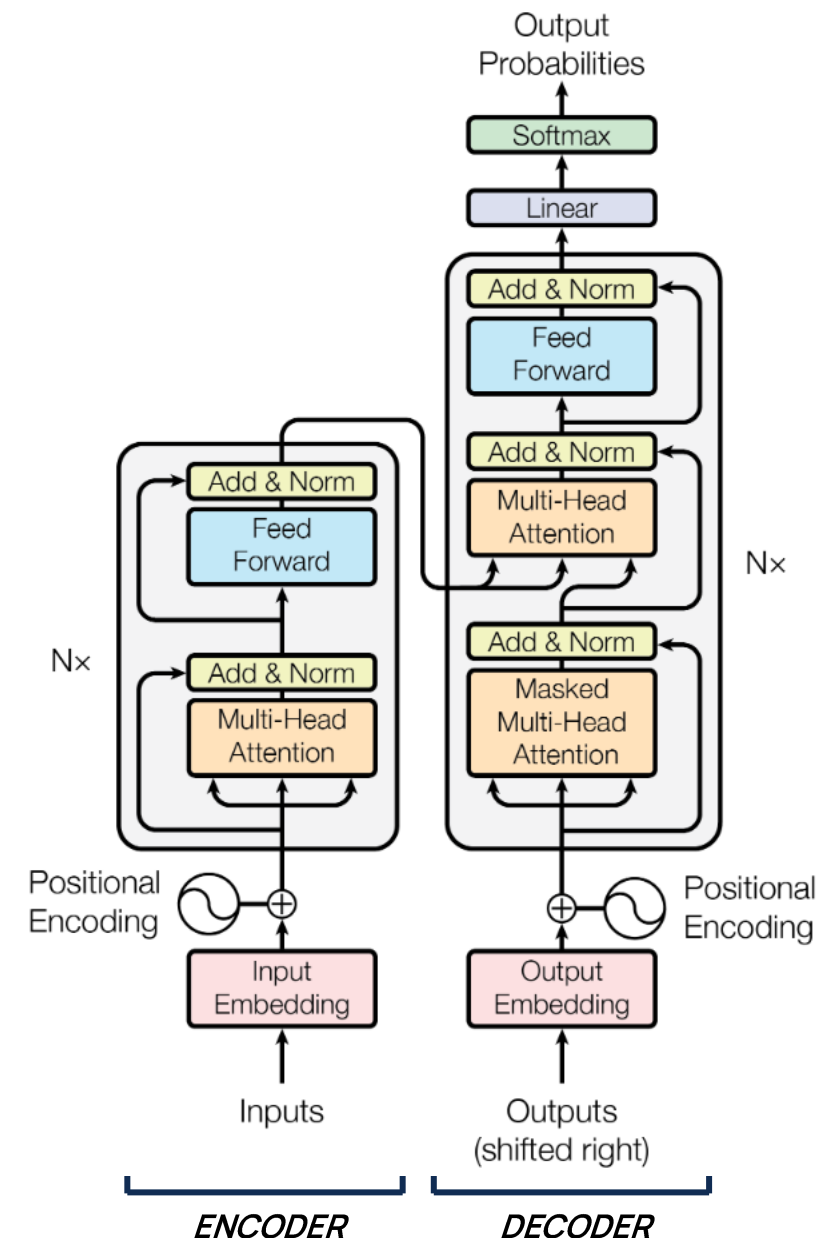
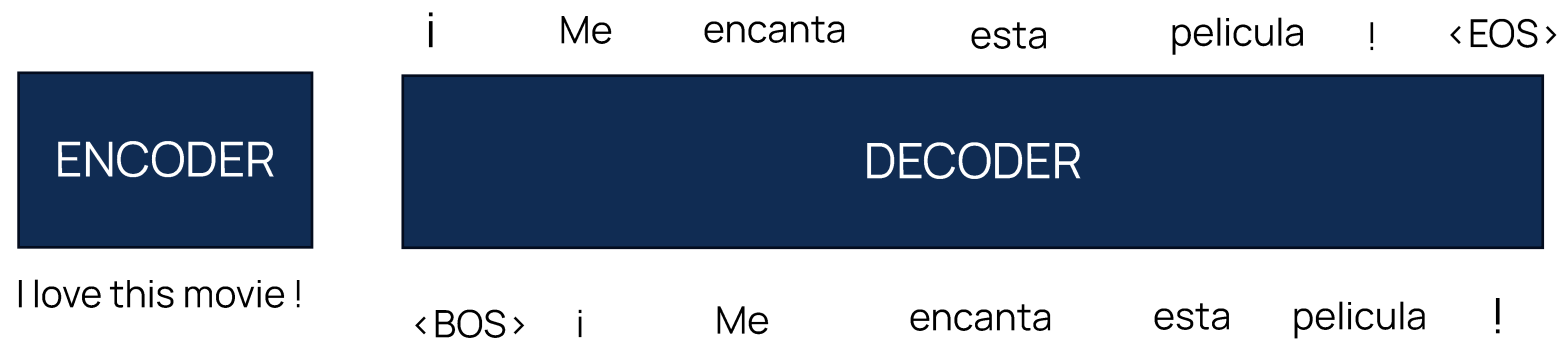


NLP 101 – Transformers (II)

The **encoder** processes all source tokens in parallel and are looking at each other

During **training** the decoder receives immediately the **entire target sequence**, shifted right and with a special <BOS> token at the beginning, allowing parallelization.

At **inference** time, the decoding process is sequential, starting with the <BOS> token only and then adding one after each other the tokens that the decoder produces

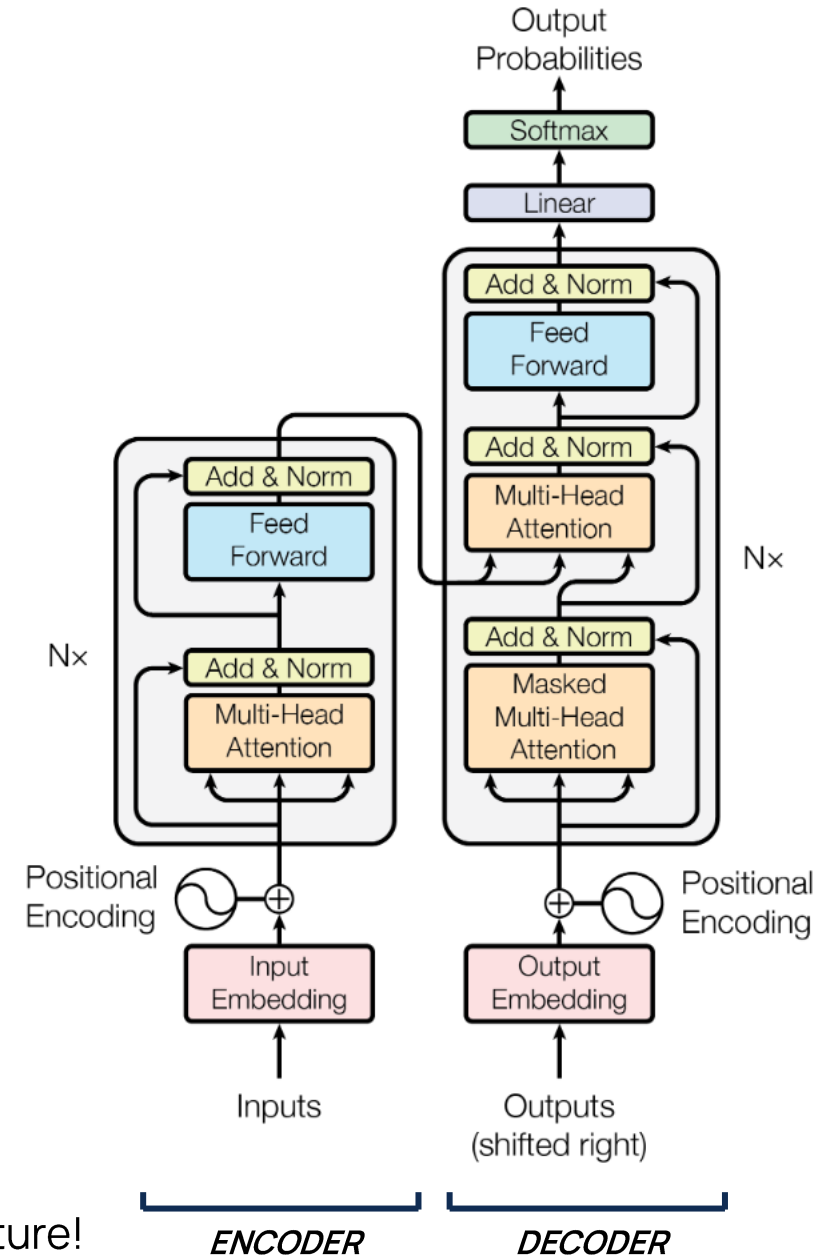


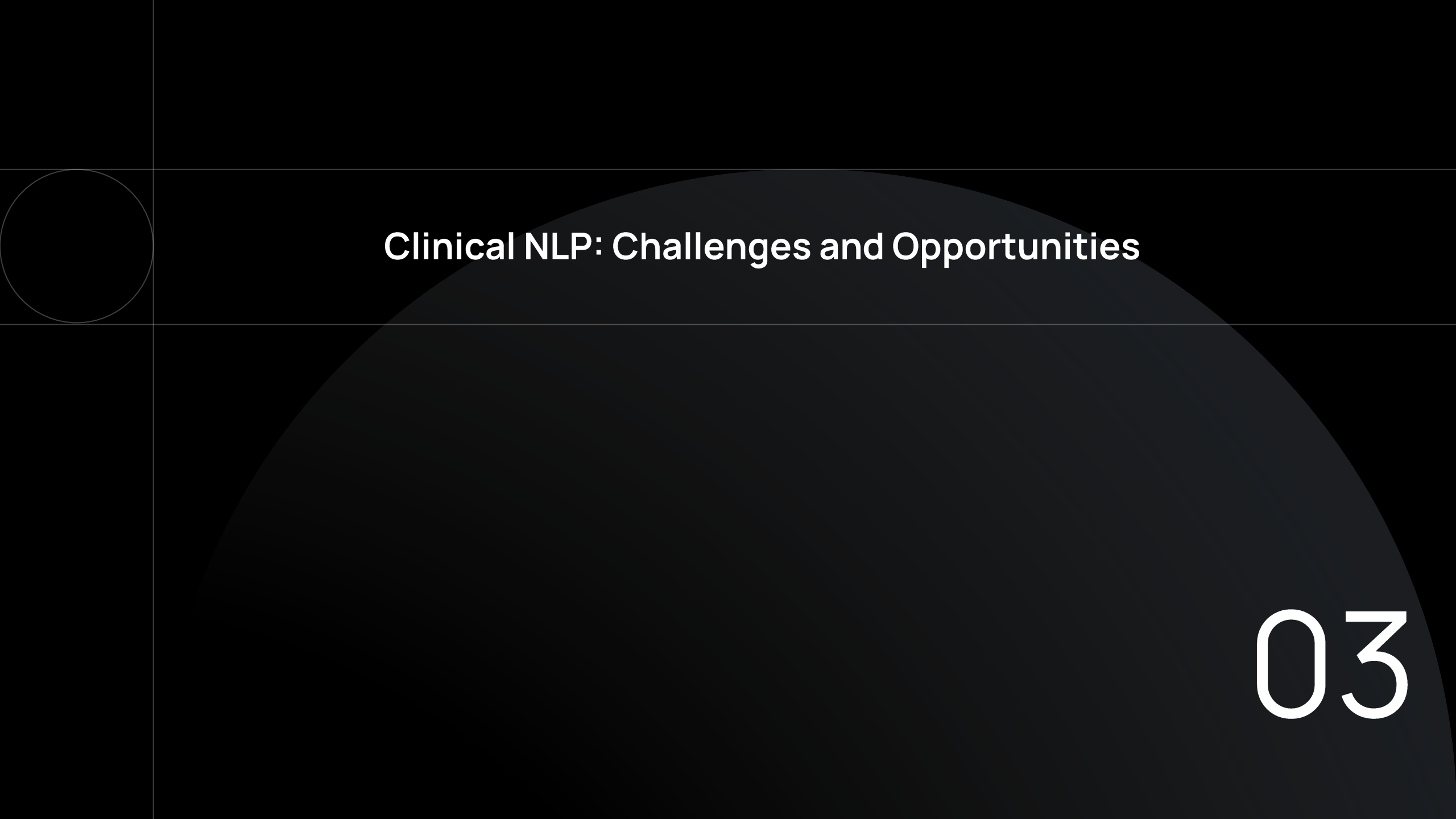
NLP 101 – Transformers (III)

ONE ARCHITECTURE, THREE PHILOSOPHIES

Encoder-Decoder	Text-to-text in the original Transformer spirit	T5 family
Encoder-only	Contextual embeddings to be used for end tasks (classification, regression, ...)	BERT family
Decoder-only	Text-to-text in a simplified architecture	GPT family

→ All modern LLMs (ChatGPT, Gemini, LLaMA, ...) are based on this architecture!





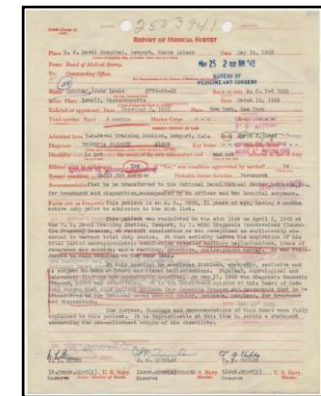
Clinical NLP: Challenges and Opportunities

03

Clinical NLP - Medical Documents

Electronic Health Records (EHR) store a large amount of information in textual format:

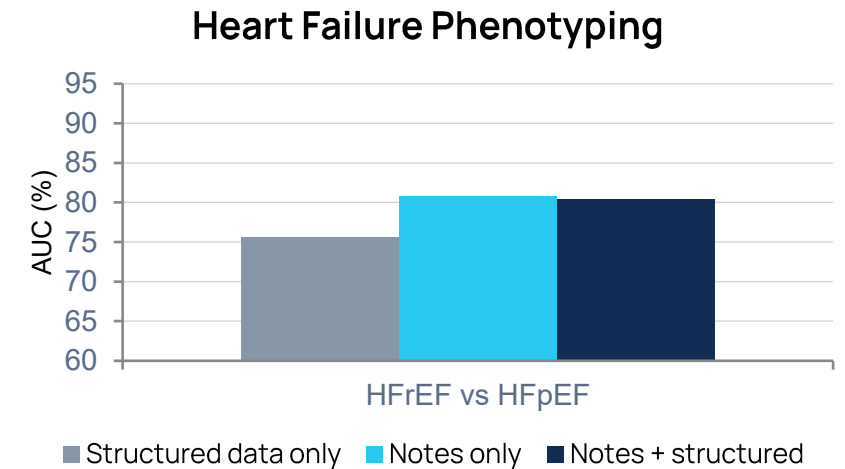
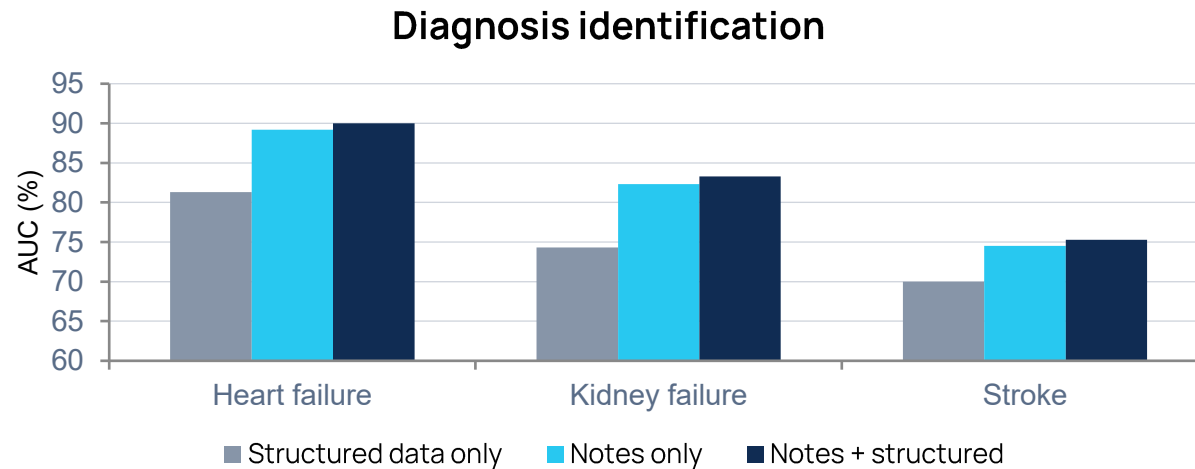
- Discharge letters
- Progress notes
- Referrals
- Outpatient visits reports
- Exam reports



Clinical NLP - Opportunities

Many information are reported **only in unstructured documents**:

- Faster to write
- No need to remember complex coding systems or interact with complex forms
- Allow the highest level of detail
- Allow the highest level of uncertainty



Clinical NLP – Challenges (I)



Domain specific, noisy and ambiguous language

Medical language is very different from traditional language, with a very specific jargon



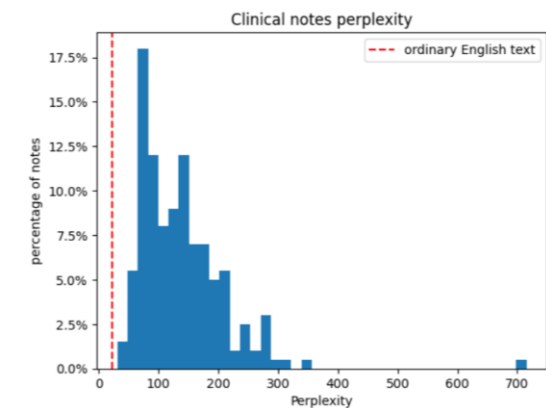
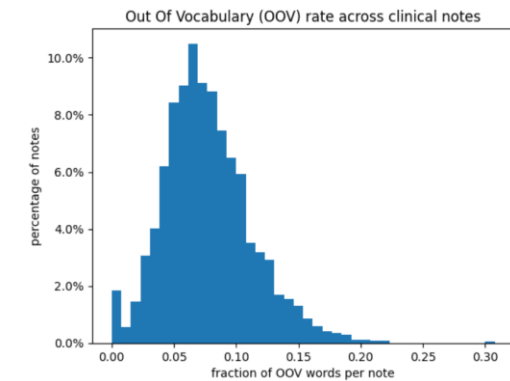
Hospitals and healthcare systems heterogeneity

There are almost no standards for medical documents



Privacy and data sharing constraints

Getting access to unstructured medical textual data is often difficult due to the higher privacy risks



Clinical NLP – Challenges (II)



Scarcity of (labelled) data

NLP models require huge amounts of data, in particular labelled data for complex, domain-specific tasks



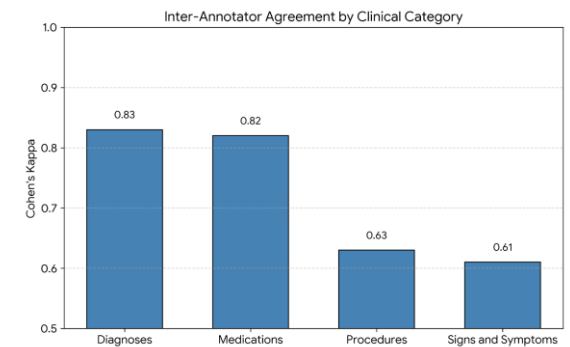
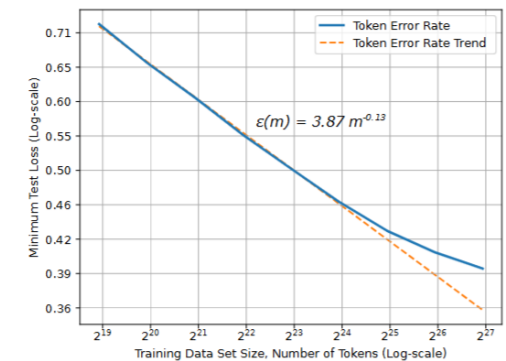
Need for interpretability

Most NLP models are black-boxes, while interpretable models are often too weak

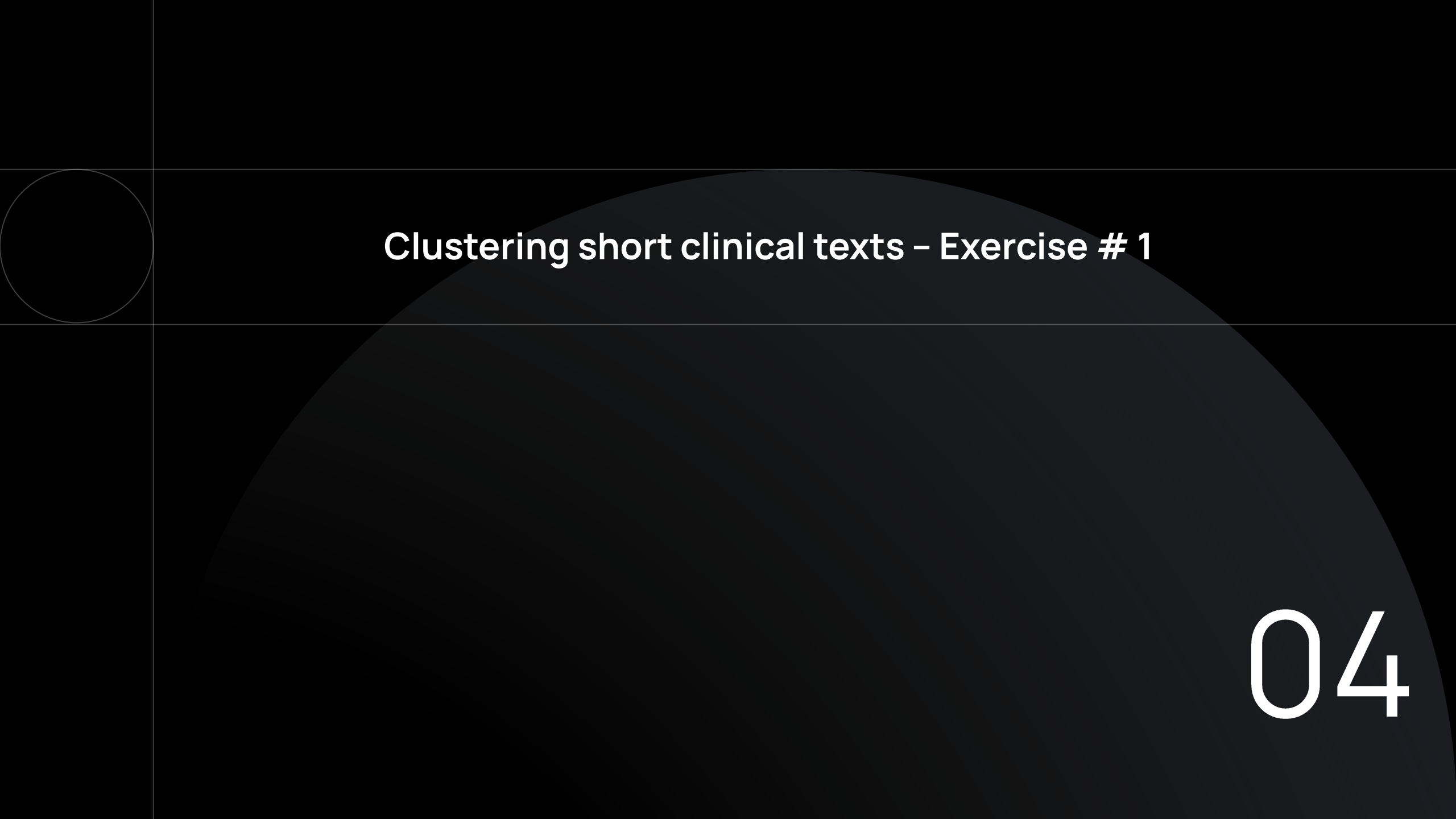


Computational limitations in clinical environments

Modern LLMs have high computational costs (GPUs)



Source: South, Brett R., et al. 'Developing a manually annotated clinical document corpus to identify phenotypic information for inflammatory bowel disease.' BMC bioinformatics 10 Suppl 9 (2009): S12.



Clustering short clinical texts – Exercise # 1

04

EX # 1 – Clustering short clinical texts

EHR often contains **short texts**:

- Unstructured fields explaining reasons or procedures
- Portion of text that can be isolated from longer documents
- Referrals
- ...

In absence of labels, **clustering** is a good option to start

Clustering long documents can be very challenging (even by hand!), while short text have more compact meaning

Diagnosis of recurrent syncope in disease of the infranodal conduction pathways

Diagnosis: continuation of treatment for acute respiratory failure during a respiratory infection in a premature twin infant with severe bronchopulmonary dysplasia.

Description
Fever in UTI

EX # 1 – Example from a real case study (I)



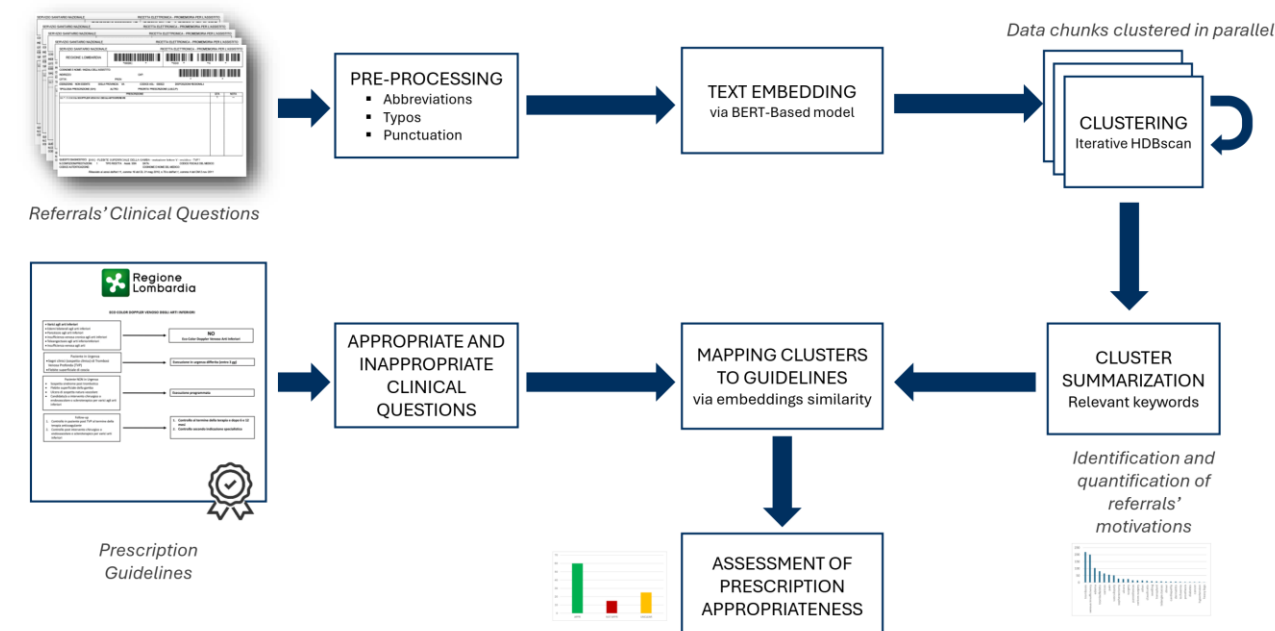
~400k + 500k referrals for two examinations (ECD and FEC)



Identify referral reasons and appropriateness against guidelines



Manual annotations on two subsets of 1,000 records each



Examples of clinical questions from referral, with English translations:

Venous insufficiency

▪[000] - in anamnesi insuff venosa arti inferiori già sottoposta a terapie sclerosanti. riferisce dolore
▪INSUFF.VOSA SAFENA ARTO INF.SX
▪insuff venoso linfatica arti inferiori prevalente arto inf sx in obesità cronica

▪[000] - in history of lower limb venous insufficiency already undergoing sclerosing therapy. reports pain
▪VENOUS INSUFFICIENCY SAFENA LEFT LOWER LIMB
▪lymphatic venous insufficiency lower limb prevailing left lower limb in chronic obesity

Varices

▪ VARICOSITA' DXmaggioreSX IN PZ GRAVIDA
▪ [000] - Follow-up di s.varicosa arti inferiori.
▪ varici superficiali gamba dx

▪RIGHT greater LEFT VARICOSITIES IN PREGNANT PATIENT
▪[000] - Follow-up of varicose leg syndrome
▪superficial varices right leg

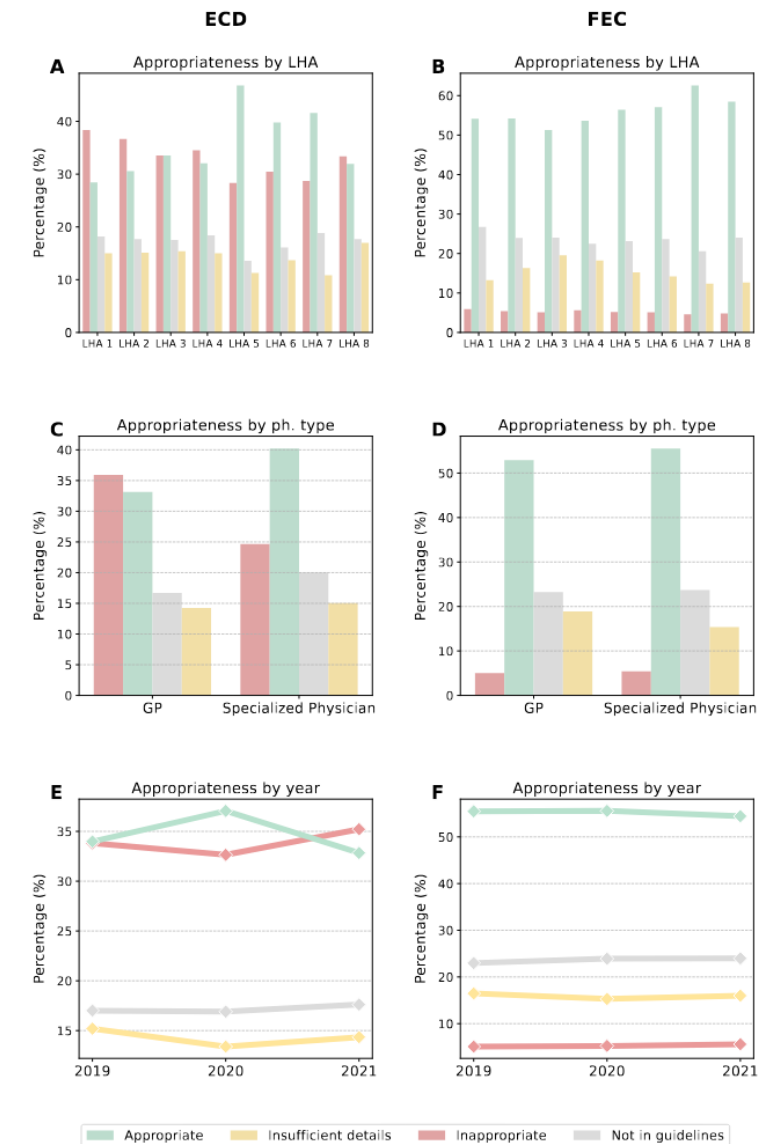
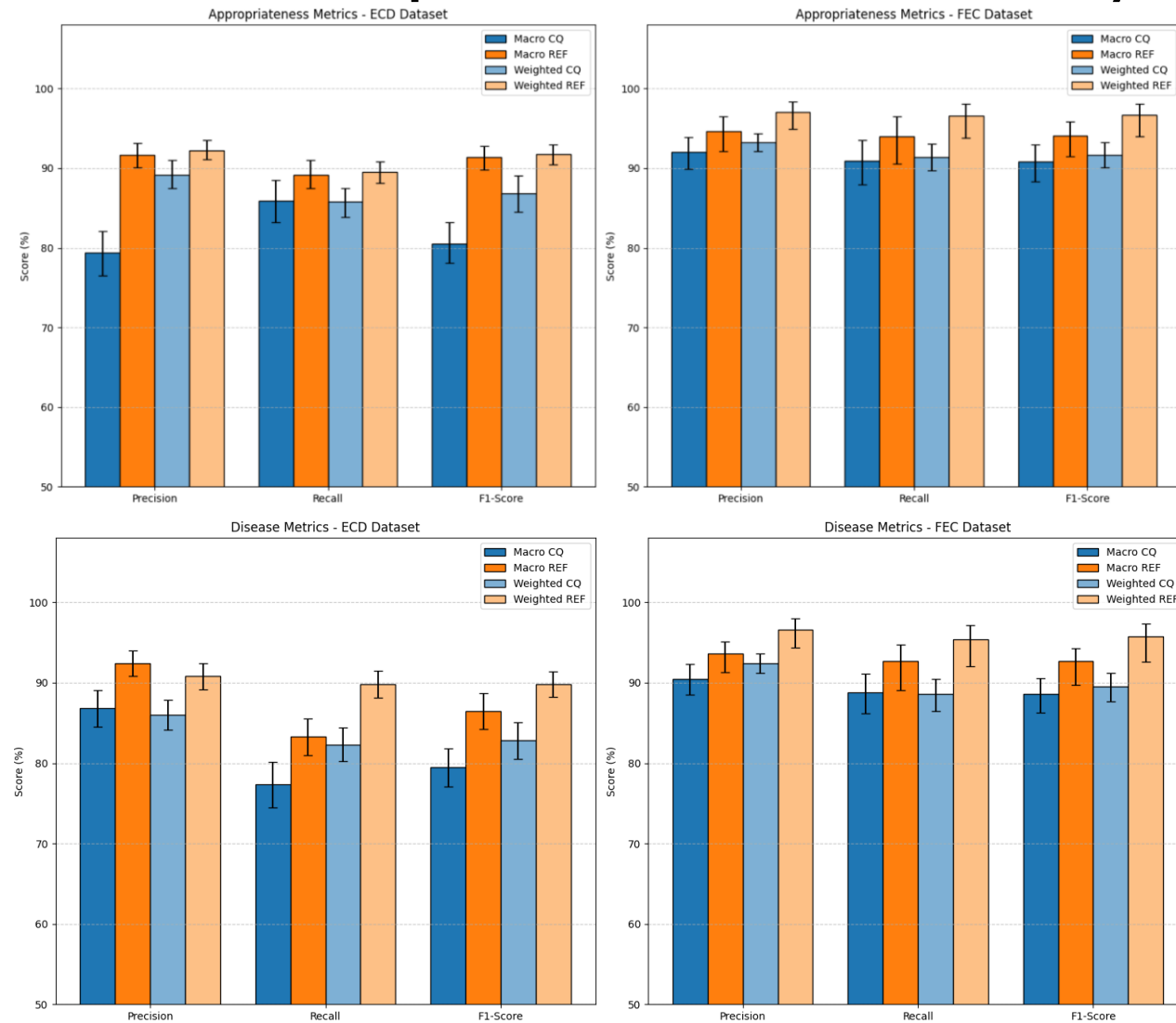
Thrombosis

▪ SOSPETTA TVP AI DESTRO
▪ Sospetta TV arto inf sx in trobocitemia con mutazione JACK2
▪ PAZIENTE CON PREGRESSA TVP FEMORALE COMUNE ARTO INFERIORE SX; RECIDIVA???

▪SUSPECTED DEEP VEIN THROMBOSIS RIGHT LOWER LIMB
▪Suspected VENOUS THROMBOSIS left lower limb in trobocythaemia with JACK2 mutation
▪PATIENT WITH PREVIOUS DEEP VEIN THROMBOSIS COMMON FEMORAL LEFT LOWER LIMB; RECURRENCE???



EX # 1 – Example from a real case study (II)



EX # 1 – Today's main dataset

“Dyspnea” admission notes at Emergency Department of San Giovanni Bosco Hospital (Turin, Italy)

- 2,667 unlabelled documents
- Real, anonymized medical data!
- Publicly available
- English version – manually and automatically translated from Italian

Released for the *CRF:filling* shared task @CL4Health – LREC 2026

Example document from the dataset (EN version)

TRIAGE: for one week, fever, productive cough, and influenza-like symptoms. At pre-triage T 38.2°C; has not taken antipyretic. This morning accidental fall as he reports slipping while sitting down on the stool. Denies head trauma, denies other complaints and symptoms.

VACCINATED 4 DOSES

MEDICAL EVALUATION: Influenza-like symptoms for about 4 days with fever and productive cough, marked asthenia and diffuse bone pain. Today fall to the ground with sacral trauma while attempting to sit on a chair; the patient reports decreased vision and therefore attributes the fall to misperception of the chair next to him. However, he was then unable to get up but was able to activate emergency services.

PMH:

- Arterial hypertension
- Diabetes mellitus on diet therapy
- Dyslipidemia
- DDD pacemaker for syncope and paroxysmal blocks on Holter
- Glaucoma with vision loss of the right eye Previous sacrococcygeal fistula
- Previous cholecystectomy

HOME MEDICATIONS: Pritor Plus 80/12.5 mg, cardioASA, Norvasc 5 mg, allopurinol 300 mg, Cardura 4 mg + 2 mg, Torvast 10 mg, Finastid 5 mg

ALLERGIES: not known

NUM_TELEFONO PARENTE



EX # 1 – What we are going to do

«Diagnosis» sentences were extracted from a subset of notes

We will cluster them:

- Extract embeddings
- Dimensionality Reduction
- Clustering Algorithm
- Clustering Summarization

Additional experiments with:

- Different embeddings
- Effects of hyperparameters
- Baseline (non-semantic) Clustering Methods

EX # 1 – Let's code!



<https://github.com/vittot/ISCB-NLP-Course-2026>

EX # 1 – Latent Dirichlet Allocation (LDA)

For each topic $k \in \{1, \dots, K\}$:

$\phi_k \sim \text{Dir}(\beta)$ [draw distribution over words]

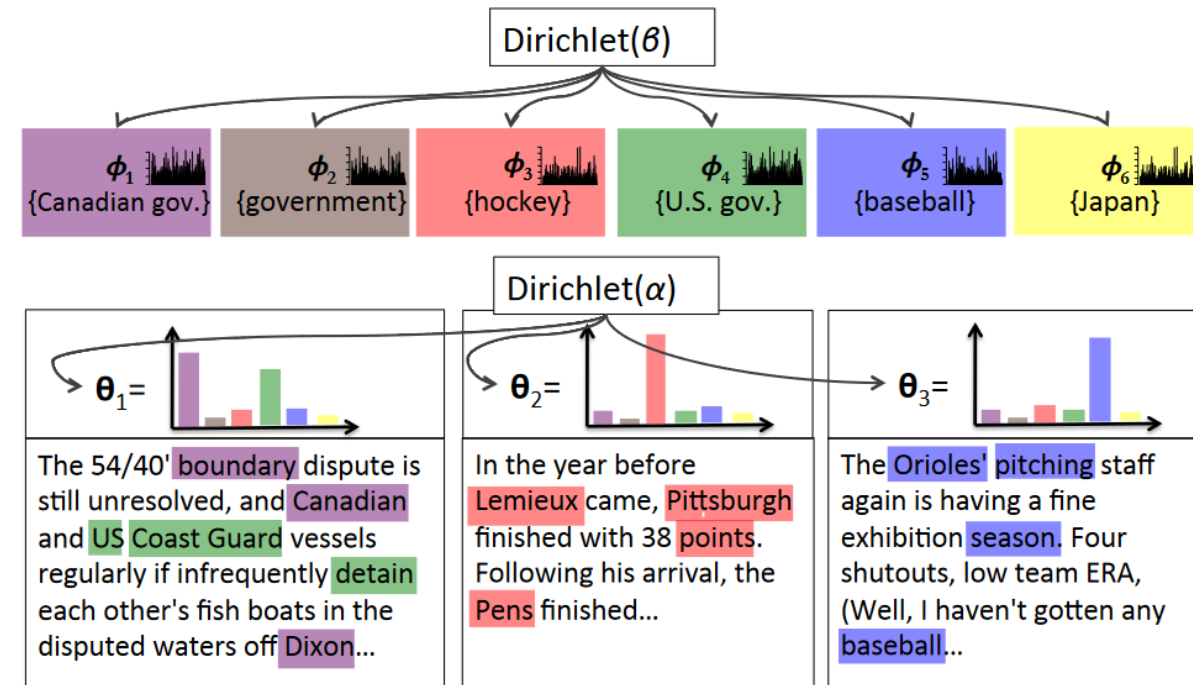
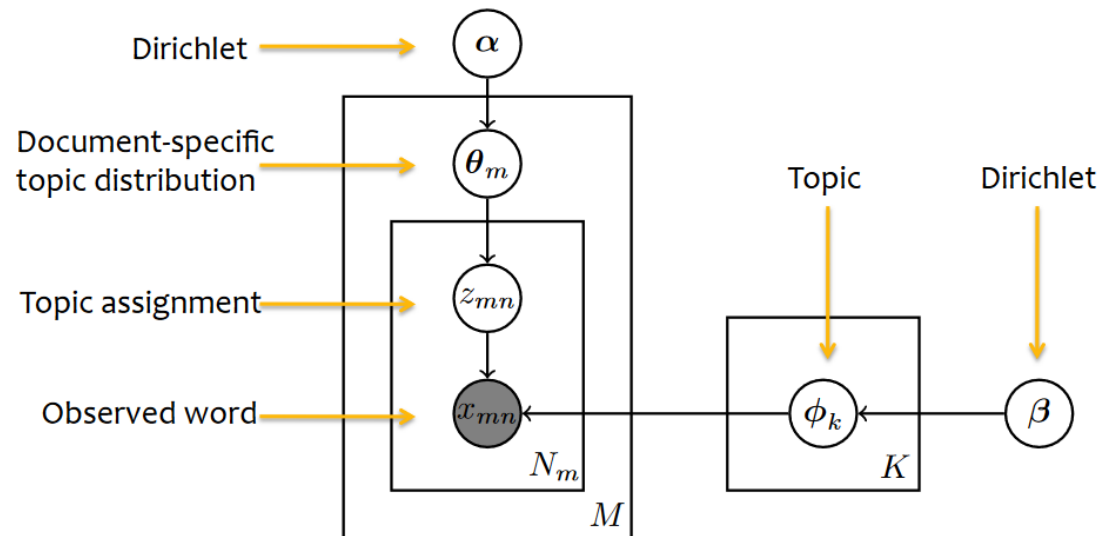
For each document $m \in \{1, \dots, M\}$

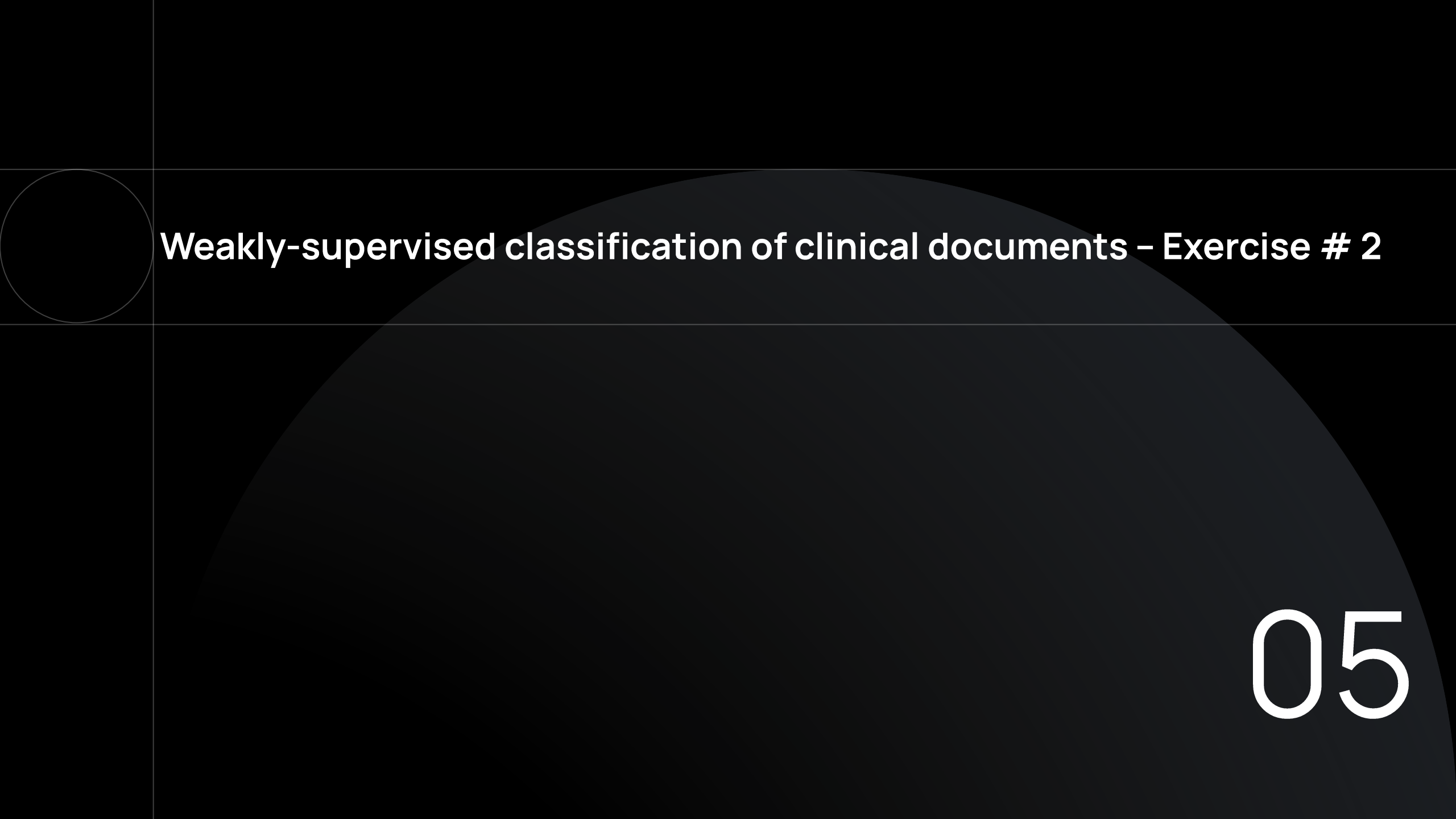
$\theta_m \sim \text{Dir}(\alpha)$ [draw distribution over topics]

For each word $n \in \{1, \dots, N_m\}$

$z_{mn} \sim \text{Mult}(1, \theta_m)$ [draw topic assignment]

$x_{mn} \sim \phi_{z_{mn}}$ [draw word]





Weakly-supervised classification of clinical documents – Exercise # 2

05

EX # 2 – Example from a real case study (I)



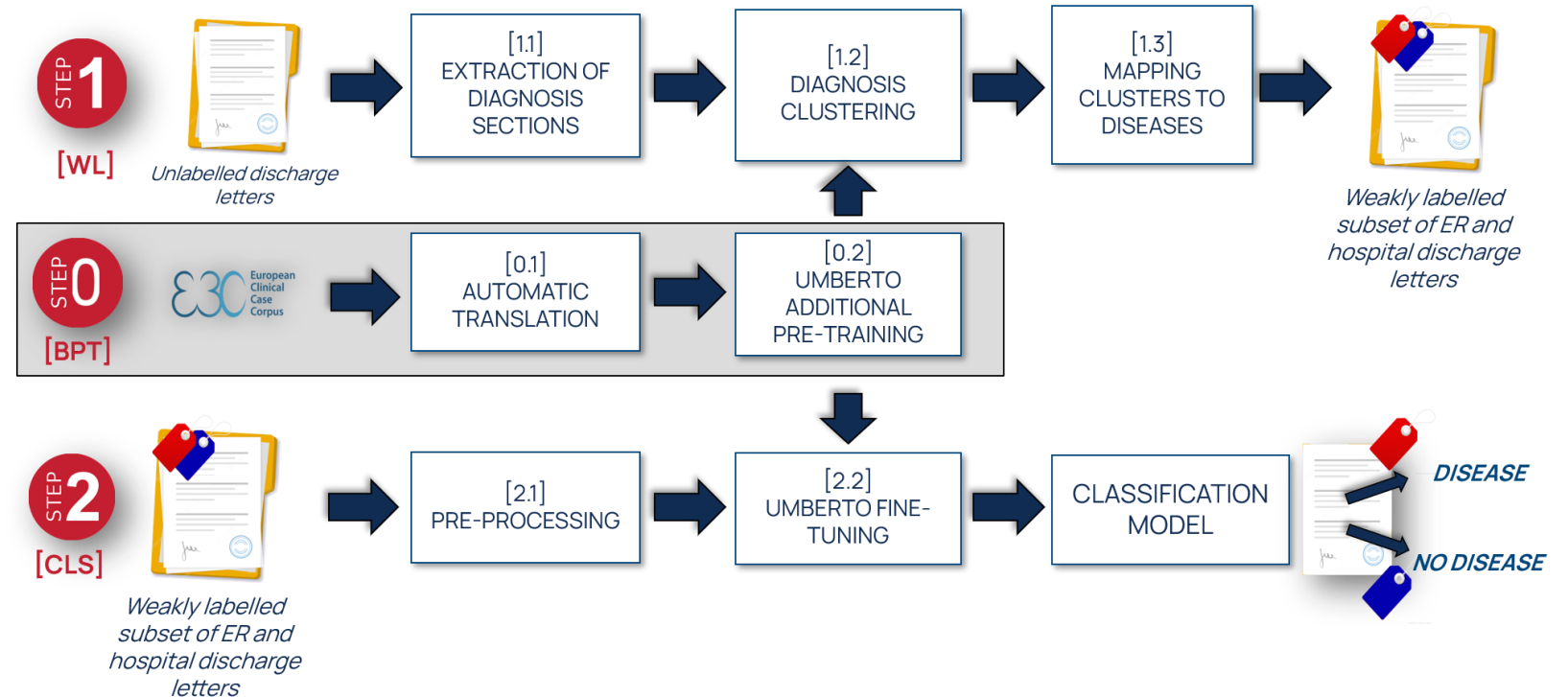
33'176 discharge letters of children admitted to ERs/Hospitals in Veneto Region between **2017** and **2019**
Collected from the **Pedianet network**



44 different hospitals, no standard template

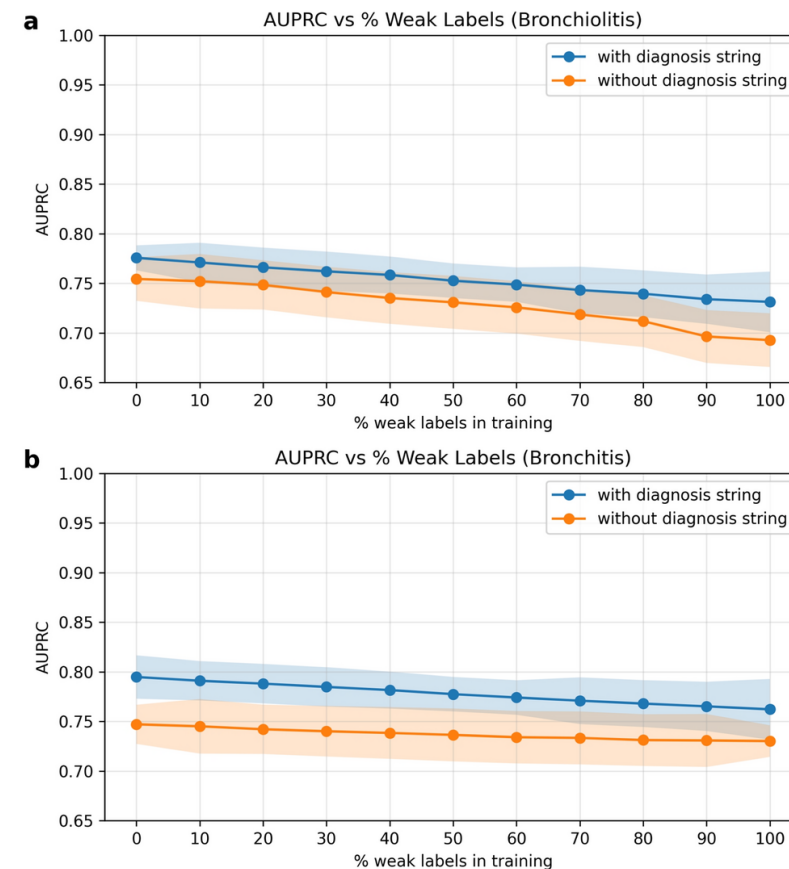
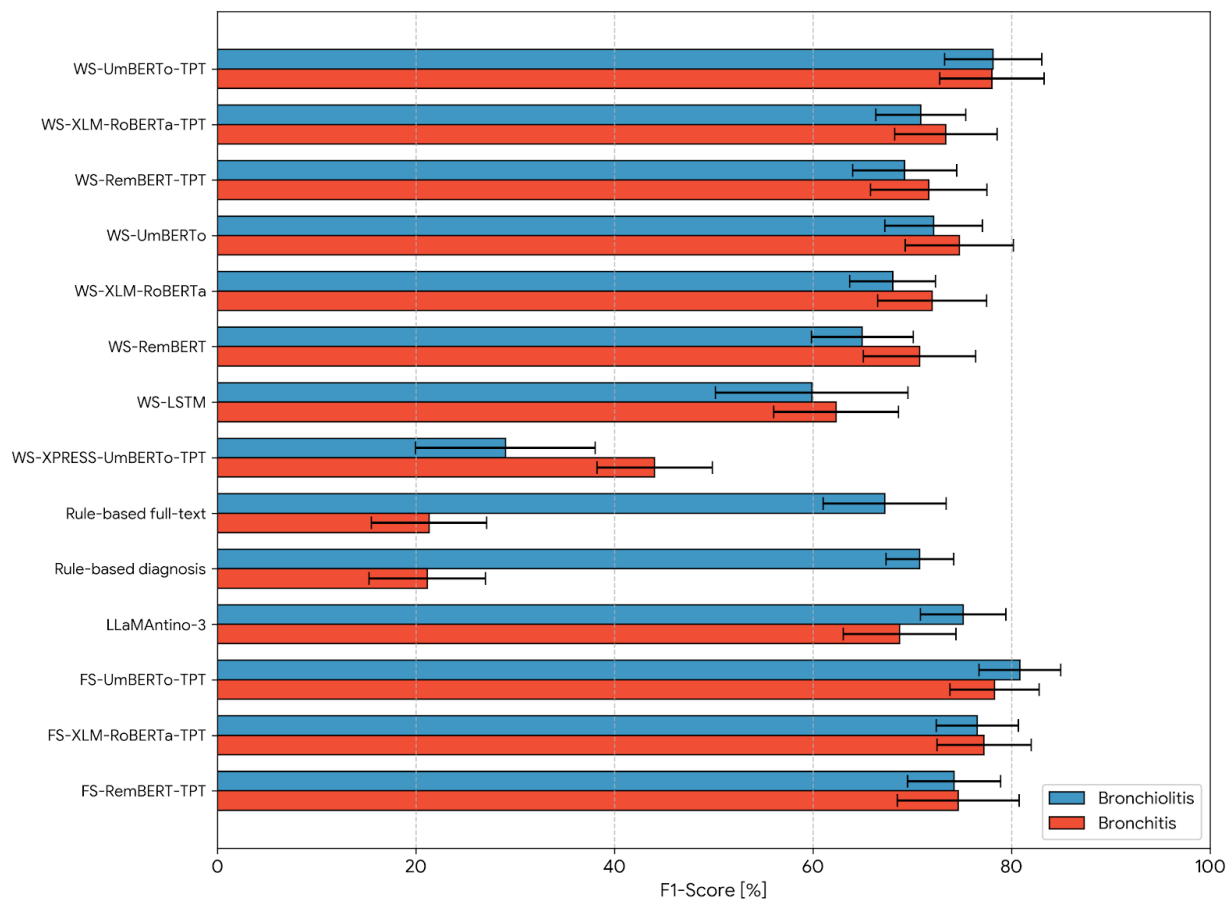


Gold labels manually annotated by clinicians for **bronchiolitis** and **bronchitis**



EX # 2 – Example from a real case study (II)

Classification Results (F1-Score) - Full Model Comparison



EX # 2 – What we are going to do

We consider the clusters obtained in EX 1:

- Convert them into “weak” labels
- Use them to train different classification models:
 - Transformer-based classifiers
 - Bag-of-Words baselines
- Assess performance
- Play with variants and hyperparameters

EX # 2 – Let's code!



<https://github.com/vittot/ISCB-NLP-Course-2026>

EX # 2 – Naive Bayes

Naive Bayes is a simple but effective document classification model

To classify a document, containing words $x_1, x_2, \dots, x_n$ we would like to estimate $P(x_1, x_2, \dots, x_n | c)$ but this would require a very large number of samples

Naive Bayes makes the **naïve assumption of conditional independency among words**:

$$P(x_1, x_2, \dots, x_n | c) = P(x_1 | c) P(x_2 | c) \dots P(x_n | c)$$

This assumption is **clearly false** in general, but despite this the model often works well for simple document classification problems

$$\hat{P}(w_i | c_j) = \frac{\text{count}(w_i, c_j)}{\sum_{w \in V} \text{count}(w, c_j)}$$

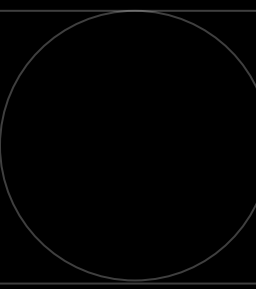
..but this would introduce issues with words that never occur in a class



$$\frac{\text{count}(w_i, c) + 1}{\left(\sum_{w \in V} \text{count}(w, c) \right) + |V|}$$

$$\hat{P}(c_j) = \frac{N_{c_j}}{N_{total}}$$

$$c_{MAP} = \operatorname{argmax}_c \hat{P}(c) \prod_i \hat{P}(x_i | c)$$



Interpretability methods for document classification

Exercise # 3

06

EX # 3 –Interpretability

Interpretability or explainability can be defined as the ability to indicate the contribution of each input feature to a model output

LOCAL INTERPRETABILITY

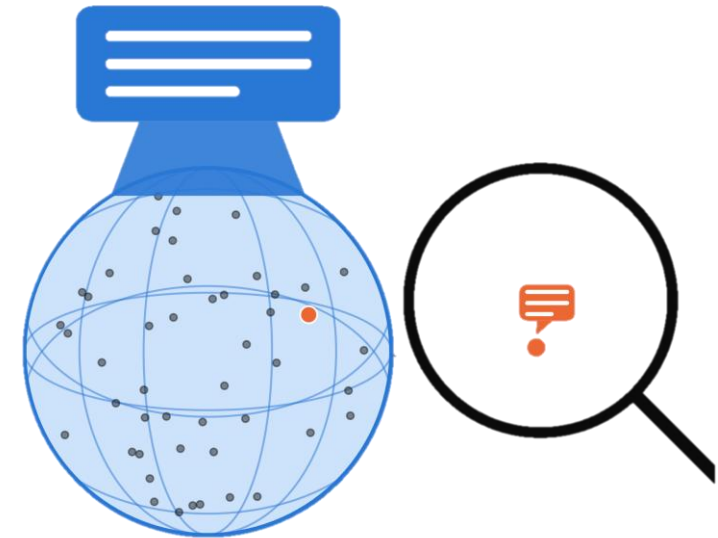
Explain individual predictions

Allows to understand predictions and biases on specific points

GLOBAL INTERPRETABILITY

Explain the model behavior “in general”

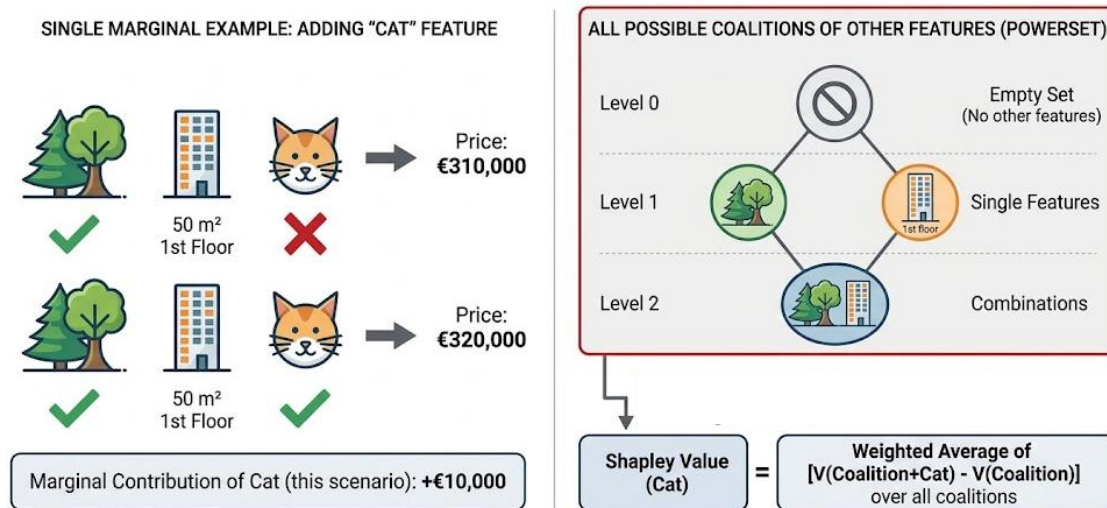
Allows to identify global reasoning and biases of the model



EX # 3 – SHAP (SHapley Additive exPlanation)

A **post-hoc, model-agnostic** explainability technique based on the concept of **Shapley Values** from Game Theory

In a nutshell: The importance of a feature j depends on the (average) output contribution it can bring to any other combination of features



To compute the importance ϕ_j of a feature j for an instance x :

Initialize $\phi_j = 0$

Repeat for $m = 1, \dots, M$:

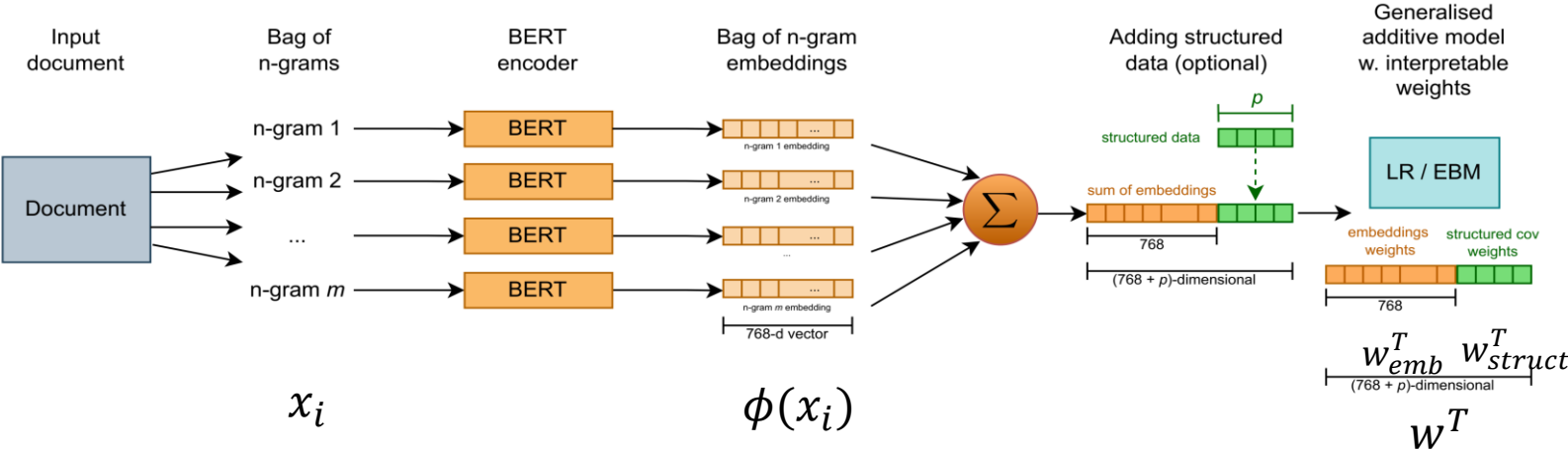
1. **Draw a random instance** z from the reference dataset (this will provide the background values to simulate missing features).
2. **Choose a random subset** S of features, ensuring you **do not include** feature j .
3. **Create instance** x_{+j} (simulating the presence of j): for the features in S and for feature j , keep the original values from instance x . For all other features, replace them with the corresponding values from z .
4. **Create instance** x_{-j} (simulating the absence of j): for the features in S , keep the original values from instance x . For all other features (including feature j), replace them with the corresponding values from z .
5. **Update** ϕ_j by adding the prediction difference (the marginal contribution):

$$\phi_j = \phi_j + \frac{1}{M} (f(x_{+j}) - f(x_{-j}))$$

EX # 3 – Example from a real case study (I)

 ~30k discharge letters of HF Patients at Amsterdam UMC (AMC + VUmc)

 Identify HF Phenotypes explaining the reasons for model outputs



LOCAL EXPLANATIONS

Each n -gram x_i and structured variable has a specific contribution to the model output:

$$score_i = w_{emb}^T \cdot \phi(x_i)$$

GLOBAL EXPLANATIONS

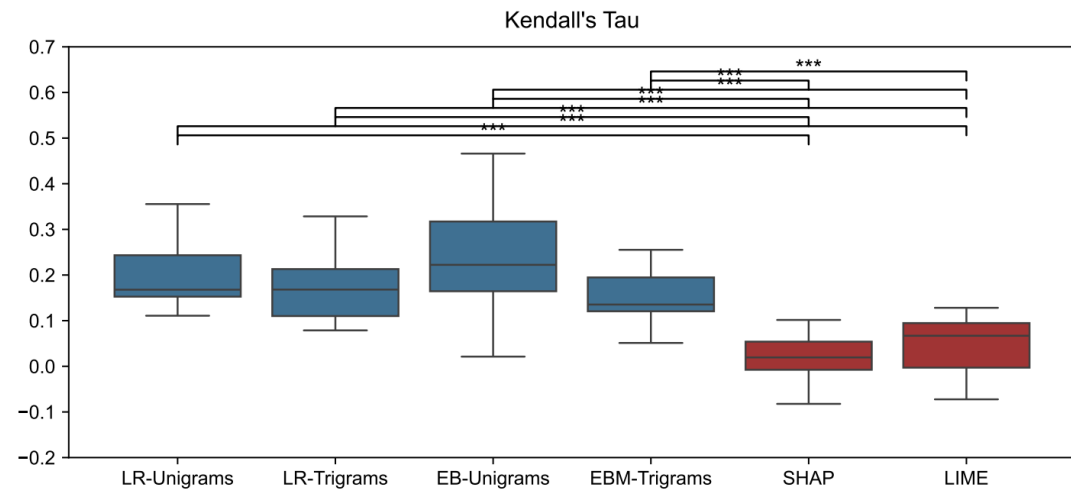
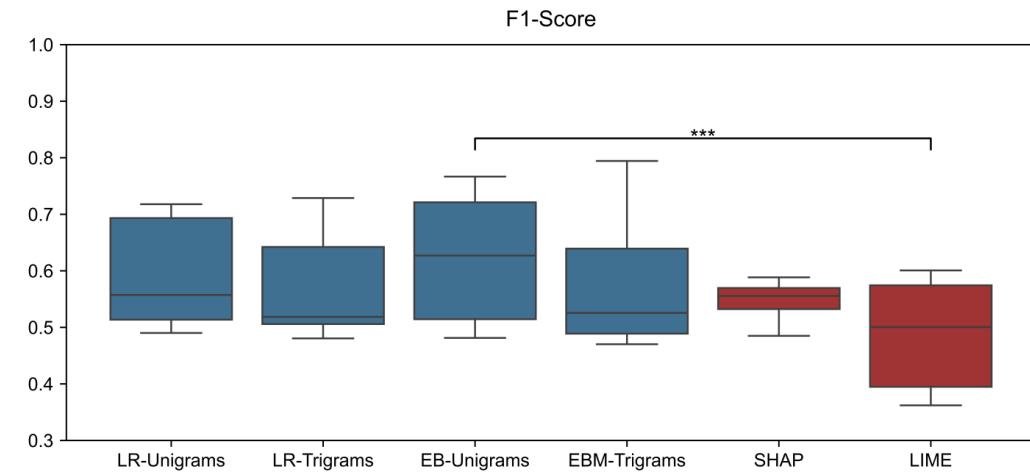
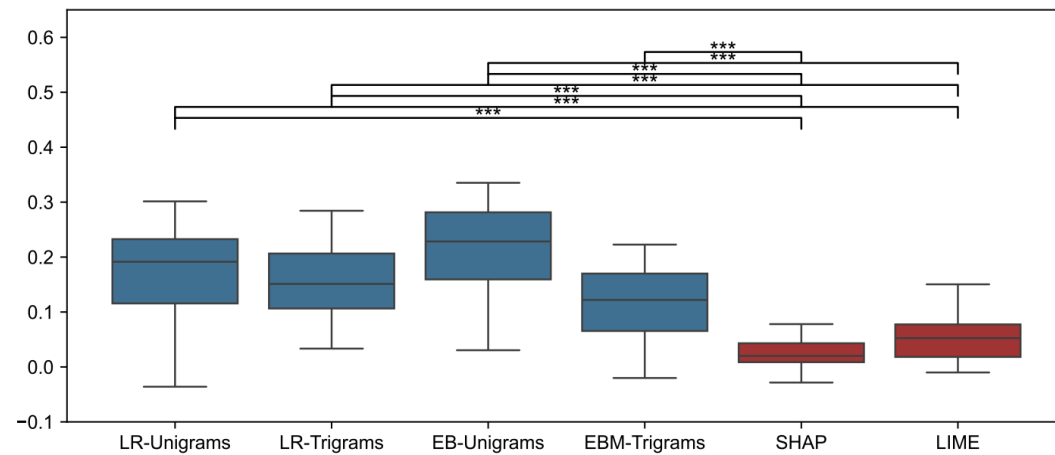
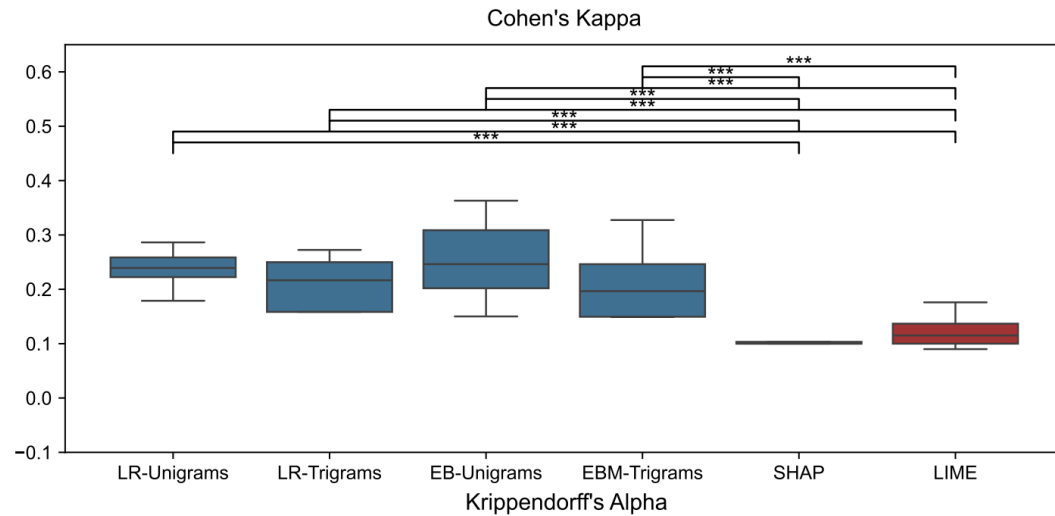
The same scores can be used as global explanations, computing scores on the whole n -grams vocabulary

 Manual annotations by two clinicians of the relevant n-grams on a subset of documents



EX # 2 – Example from a real case study (II)

INTERPRETABILITY RESULTS – AGREEMENT WITH CLINICIANS' REASONING



EX # 3 – What we are going to do

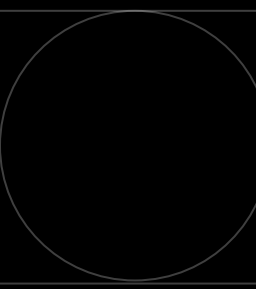
Consider the Transformer-based classification models built in **EX # 2**

1. Implement their post-hoc explanations with **SHAP**
2. Implement **Aug-Linear** classifiers
3. **Compare** their explanations

EX # 3 – Let's code!



<https://github.com/vittot/ISCB-NLP-Course-2026>



LLM-Prompting for information extraction

Exercise # 4

07

EX # 4 – Example from a real case study (I)



Original Italian version of the Dyspnea clinical notes we used in previous examples

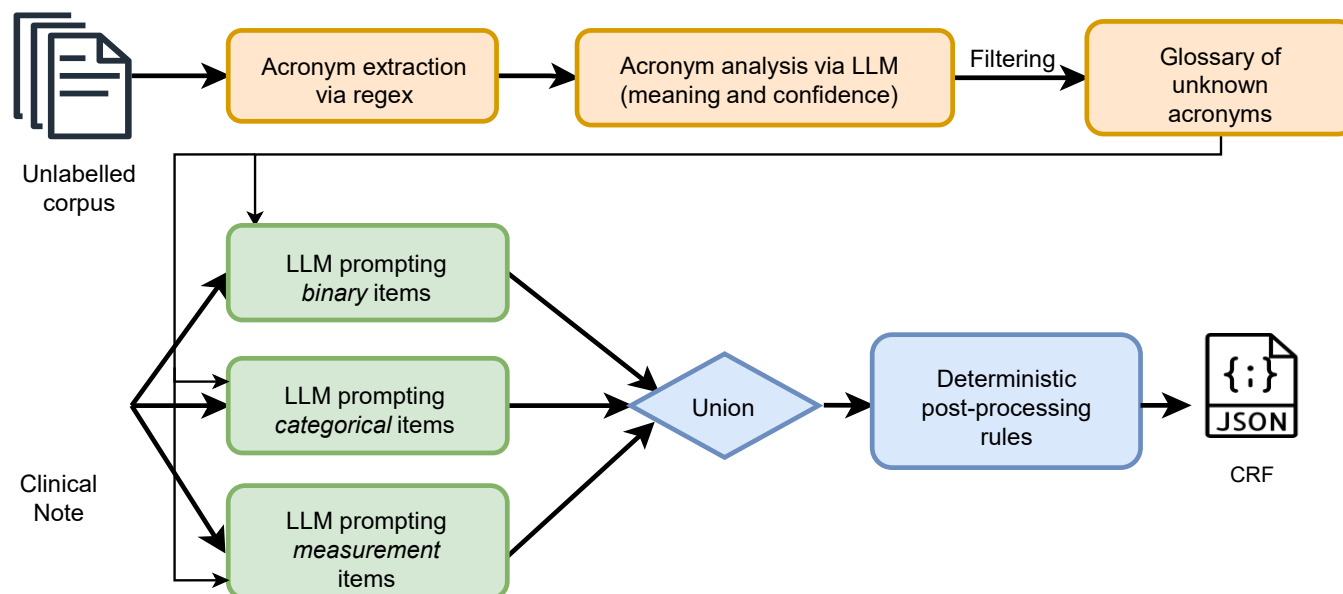


Extract 134 items:

- 4 **measurements** (*blood potassium, blood sodium, hemoglobin, ...*)
- 14 **categorical** (*blood pressure, heart rate, chronic cardiac failure, ...*)
- 96 **binary** (*aortic dissection, acute coronary syndrome, concussive head trauma, ...*)



Manual annotations on 90 notes



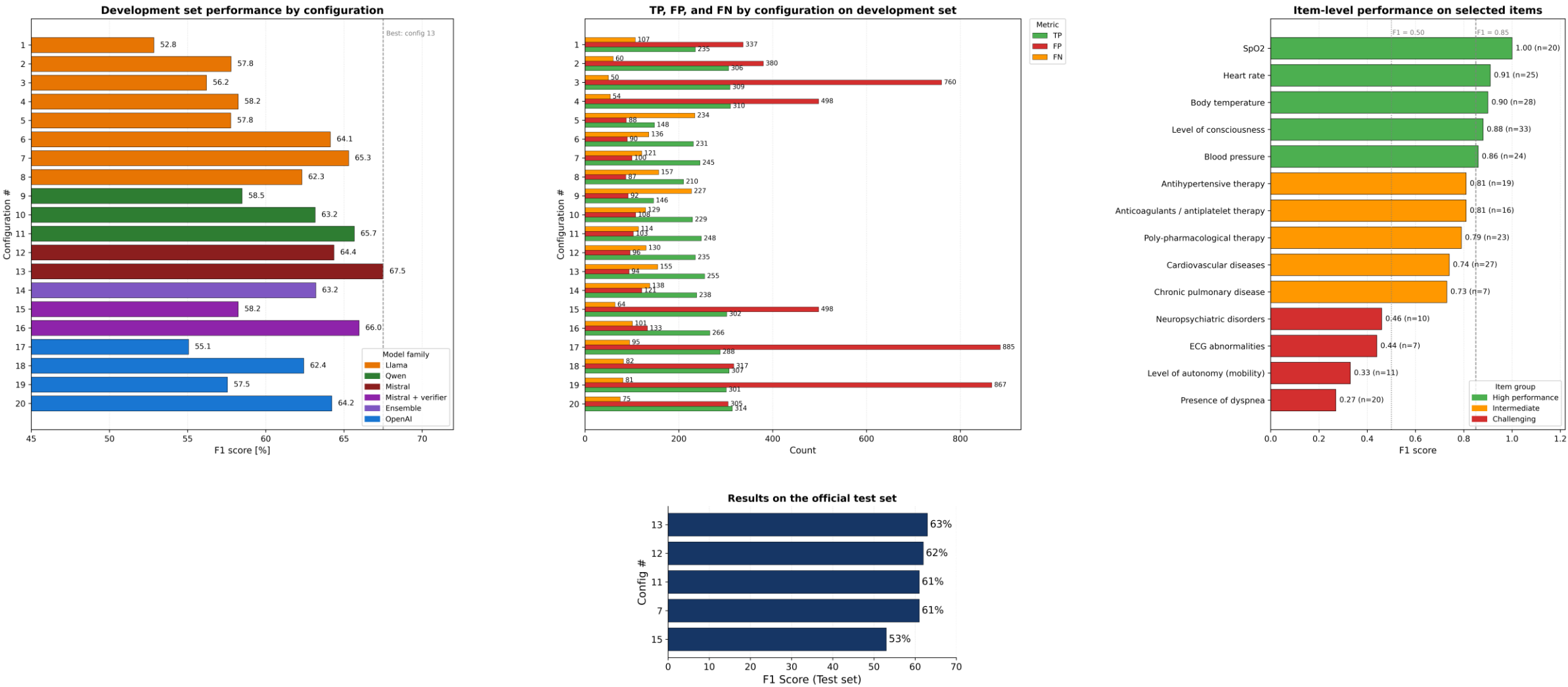
Many items ~95% of *unknown* values
Average items per CRF: 5.4 (min=0, max=30)

```

history of allergy : n
history of recent trauma : n
diffuse vascular disease : y
leukocytes : 24530
c-reactive protein: 0.69
chest rx, any abnormalities : n
  
```



EX # 4 – Example from a real case study (II)



EX # 4 – What we are going to do

We will try to replicate the process with some differences/limitations:

- English notes
- Limited computational power on Colab:

LLaMA 70B , Qwen 72B and Mistral 24 B cannot run there

Llama 3.1 8B can, using the most powerful GPU environment offered by Colab (A100)

Qwen2.5-3B works also with the smallest Colab GPU Environment (T4) – but it takes eternity to run!

EX # 4 – Let's code!



<https://github.com/vittot/ISCB-NLP-Course-2026>

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BIG DATA
small data
Your Data

Transitions in the age of biomedical AI

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Questions?



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Thank you for the attention!

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